Molecular Biology Course

Lecture 6

PROKARYOTE AND EUKARYOTE TRANSCRIPTION

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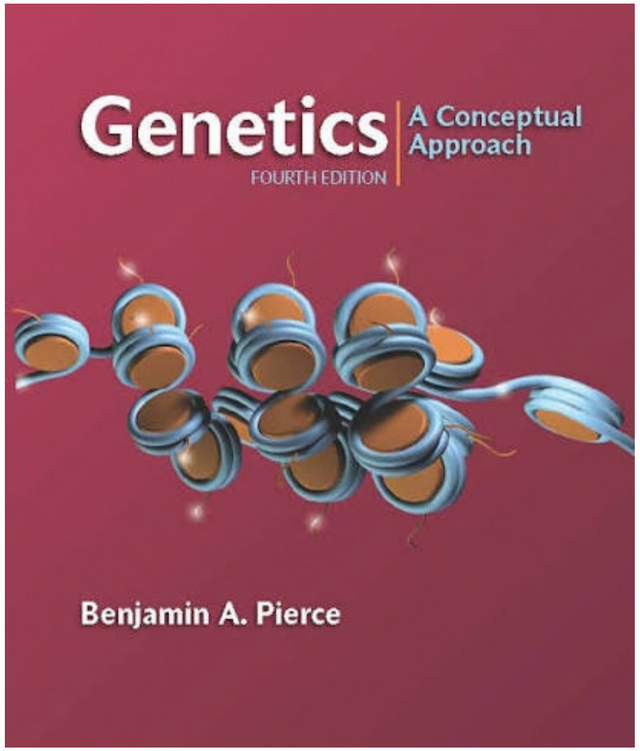
TEXTBOOK

Genetics A Conceptual Approach (Benjamin A. Pierce)

https://axonb.com/wp-content/uploads/2020/07/Genetics-A-Conceptual-Approach-by-Banjamin-A.-Pierce-4th-Edition.pdf



CONTENTS

- RNA molecule
- Transcription in Prokaryote
- Transcription in Eukaryote
- RNA processing

RNA MOLECULE STRUCTURE

Like DNA, RNA is polymer consisting of nucleotides joined together by phosphodiester bonds

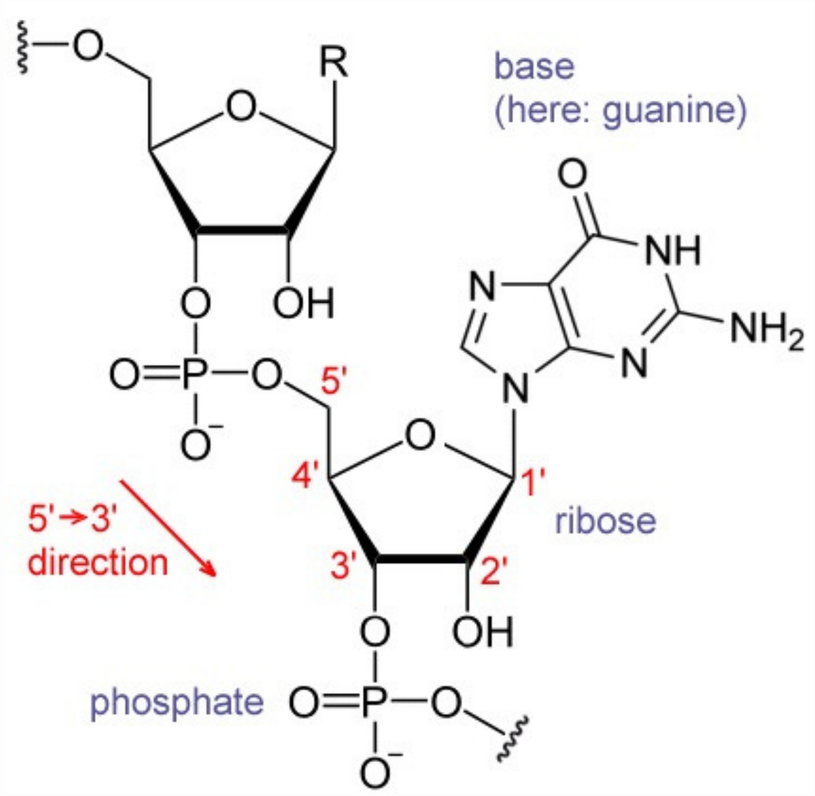
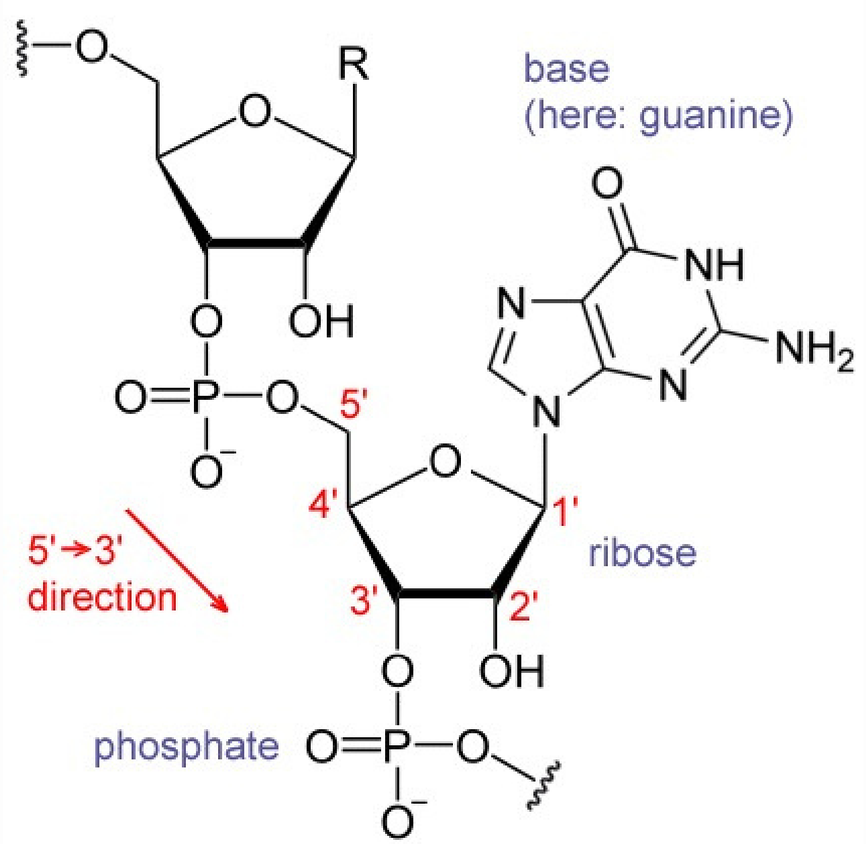


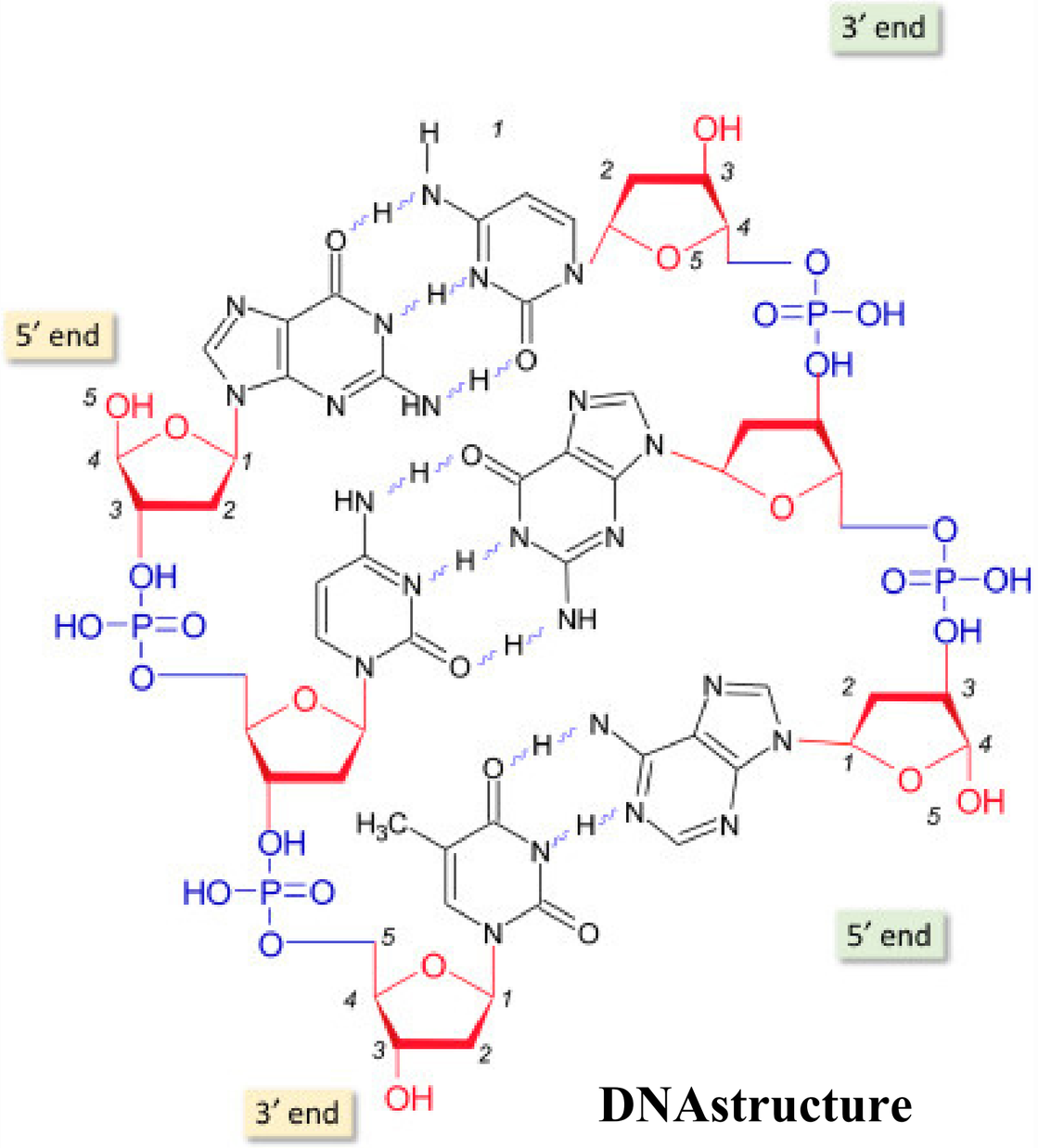
Table 13.1 The structures of DNA and RNA compared

Characteristic	DNA	RNA
Composed of nucleotides	Yes	Yes
Type of sugar	Deoxyribose	Ribose
Presence of 2′-OH group	No	Yes
Bases	A, G, C, T	A, G, C, U
Nucleotides joined by phosphodiester bonds	Yes	Yes
Double or single stranded	Usually double	Usually single
Secondary structure	Double helix	Many types
Stability	Quite stable	Easily degraded

Like DNA, RNA is polymer consisting of nucleotides joined together by phosphodiester bonds

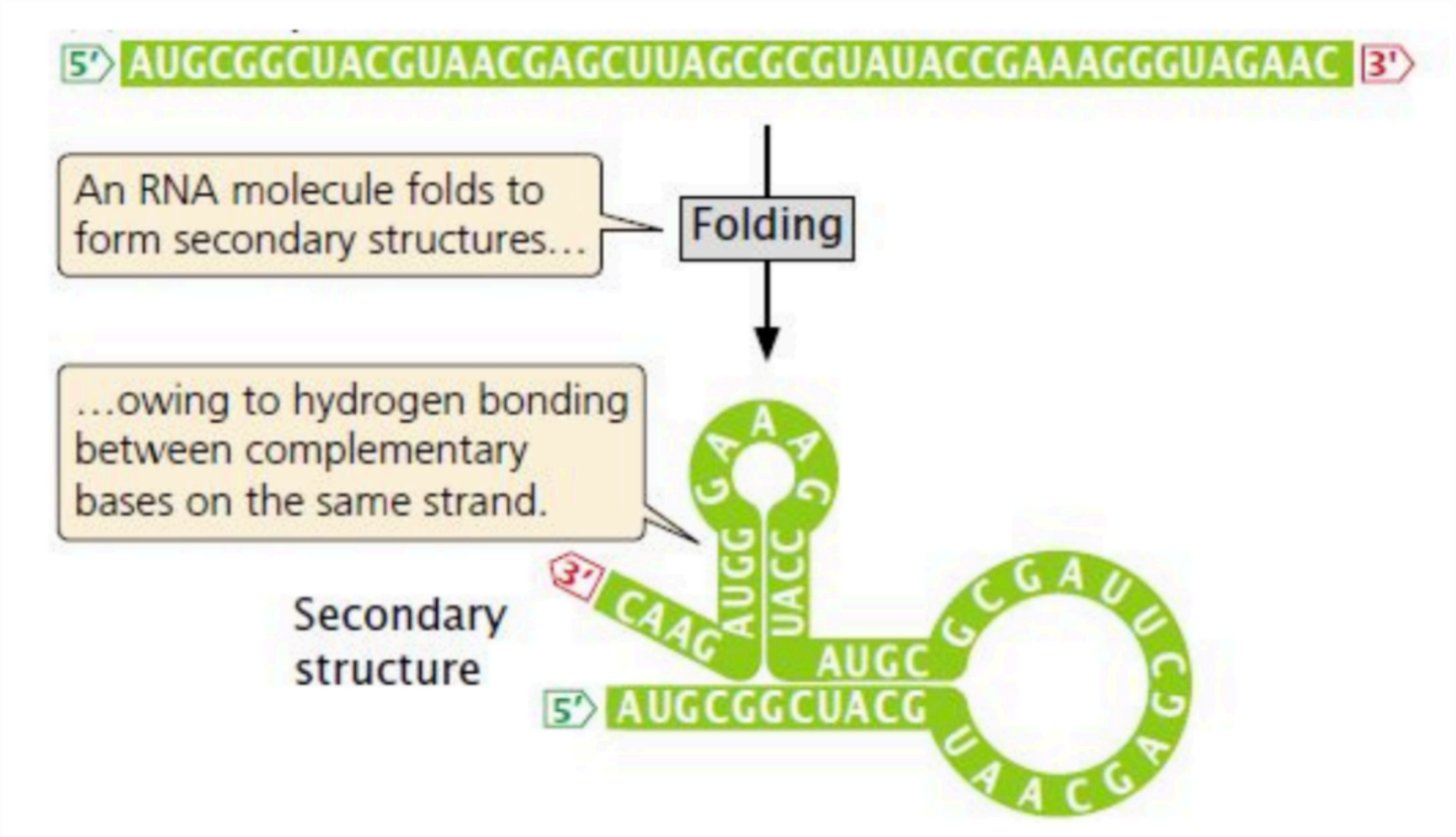


RNAstructure



DNAstructure

- The formation of secondary structure of RNA plays roles in cellular function
- Structure is easy to vary → different RNA molecules with different functions



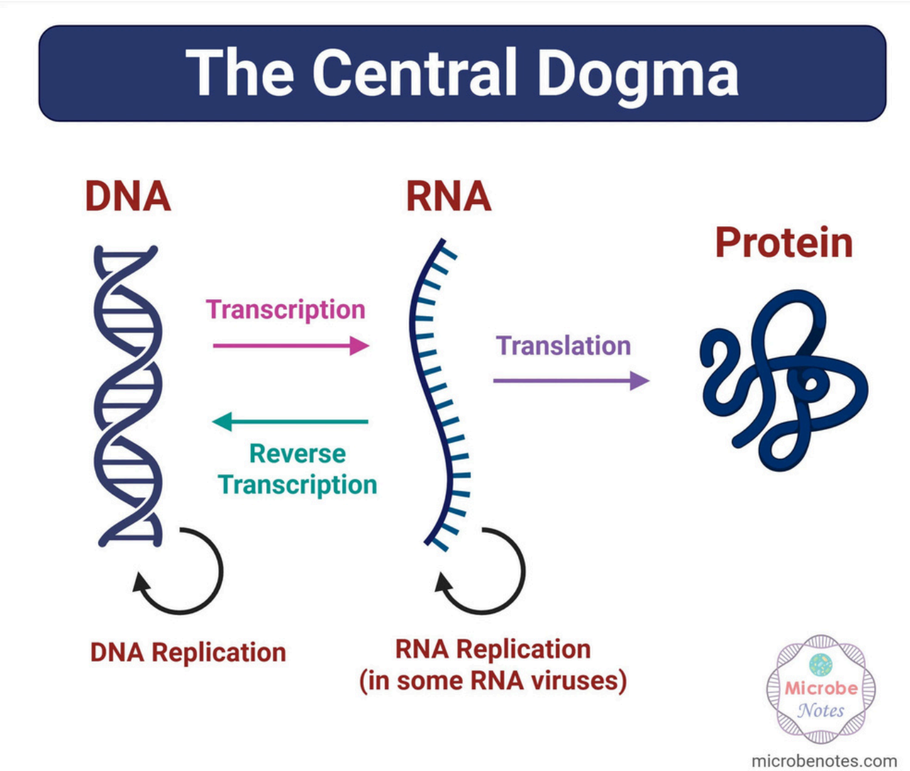
Classes of RNA molecules

Table 13.2 Locations and functions of different classes of RNA molecules

Class of RNA	Cell Type	Location of Function* in Eukaryotic Cells	Function
Ribosomal RNA (rRNA)	Bacterial and eukaryotic	Cytoplasm	Structural and functional components of the ribosome
Messenger RNA (mRNA)	Bacterial and eukaryotic	Nucleus and cytoplasm	Carries genetic code for proteins
Transfer RNA (tRNA)	Bacterial and eukaryotic	Cytoplasm	Helps incorporate amino acids into polypeptide chain
Small nuclear RNA (snRNA)	Eukaryotic	Nucleus	Processing of pre-mRNA
Small nucleolar RNA (snoRNA)	Eukaryotic	Nucleus	Processing and assembly of rRNA
Small cytoplasmic RNA (scRNA)	Eukaryotic	Cytoplasm	Variable

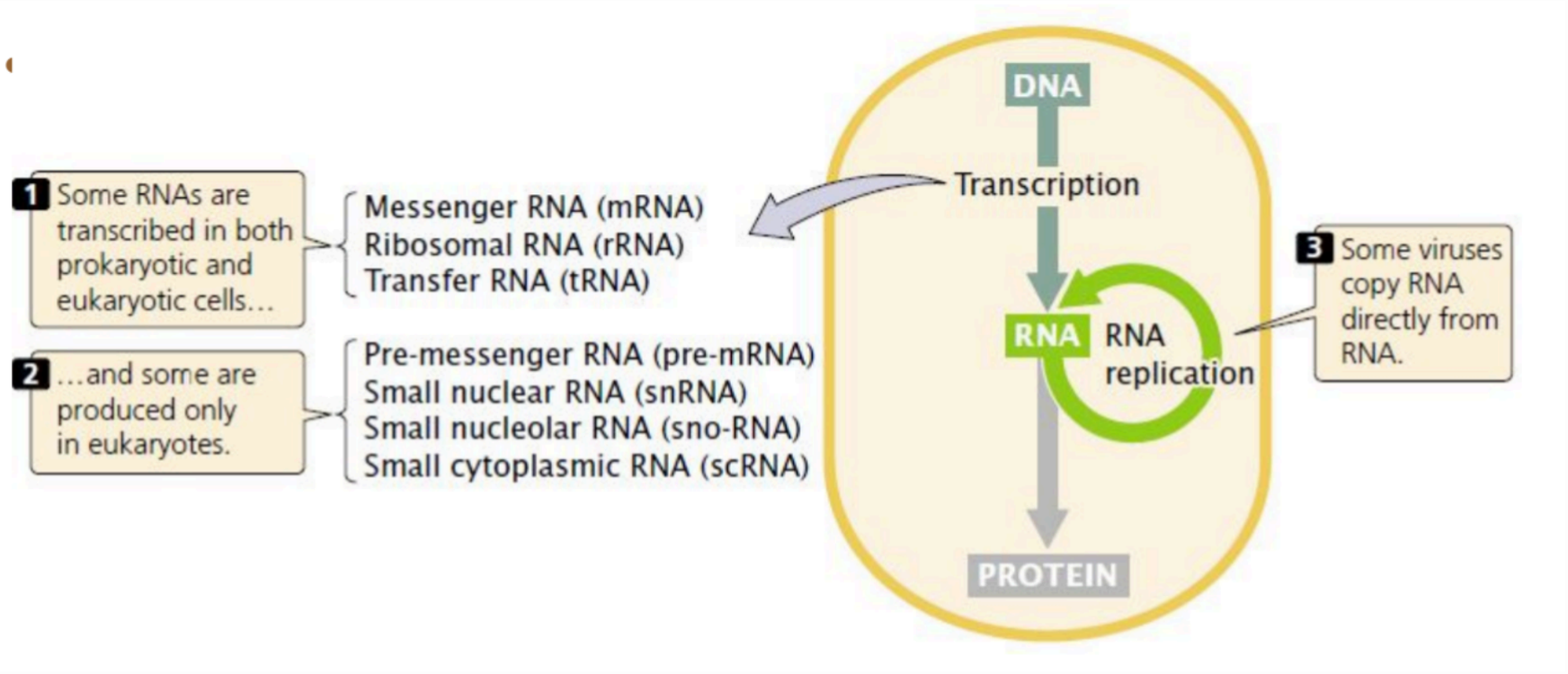
*All eukaryotic RNAs are transcribed in the nucleus.

Transcription ??



- Transcription is the synthesis of RNA molecules with DNA template
- The first step of transfer genetic information from genotype to phenotype
- Require number of proteins
- Focus on understanding how these components relate to the overall purpose of the process

- All RNA is synthesized from DNA template
- Some regions of DNA is transcribed only (not all molecules) depending on the cellular requirement



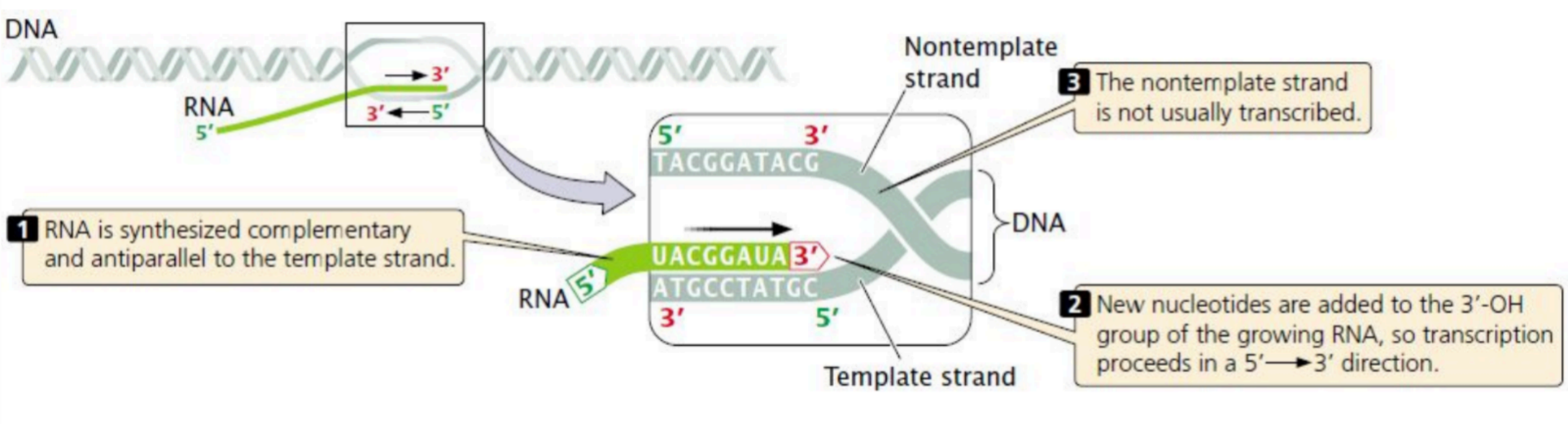
Tại sao cùng một genome, nhưng ở tế bào khác nhau lại biểu hiện gene khác nhau cho những tế bào đó ??

Process of transcription requires 3 major components

- DNA template
- Substrates needed to build new RNA molecules: nucleotides
- The transcription apparatus (protein enzymes)

https://www.youtube.com/watch?v=YlOqI3PQwjo

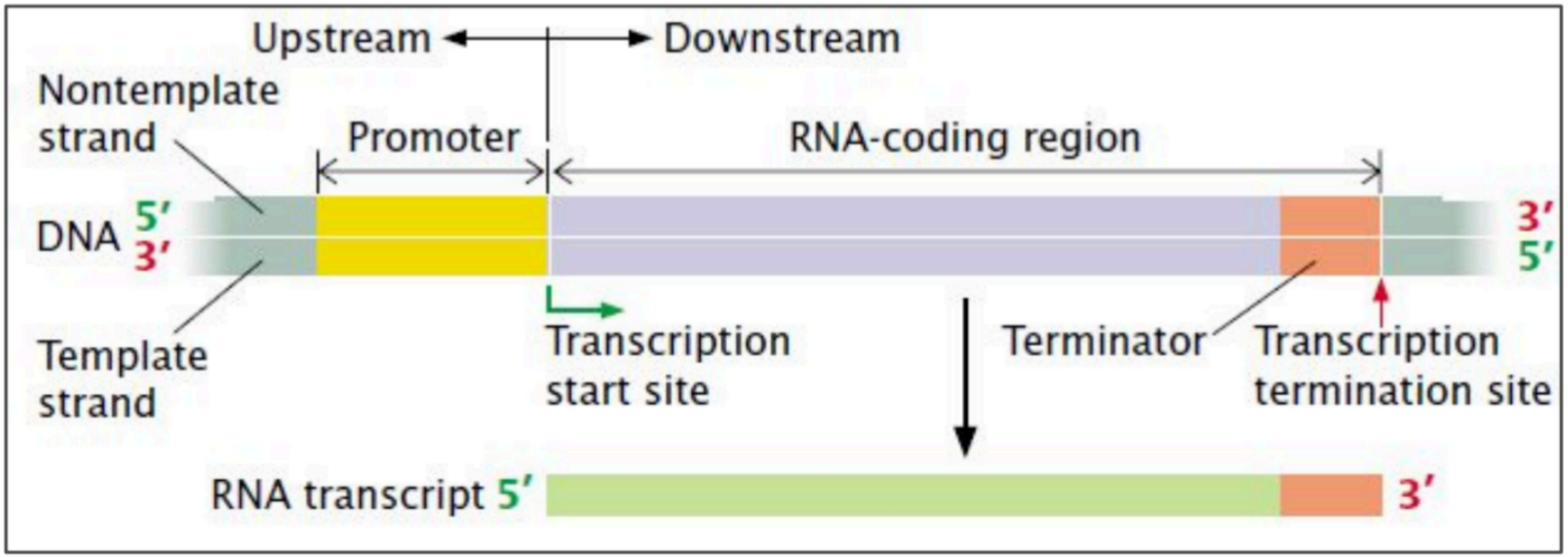
- Templates: one single strand of DNA molecules called template strand, other nontemplate strand is not usually transcribed
- The RNA product is complement with the template and has the sequence and polarity similar to the nontemplate strand (except for some nucleotides at T- replaced with U)



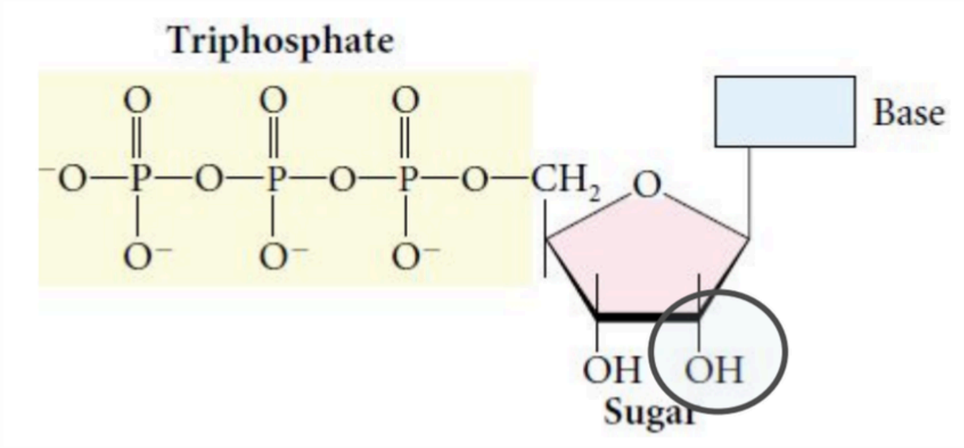
Transcription unit: the stretch of DNA that codes for RNA sequence and necessary for transcription

Included in transcription unit are 3 critical components:

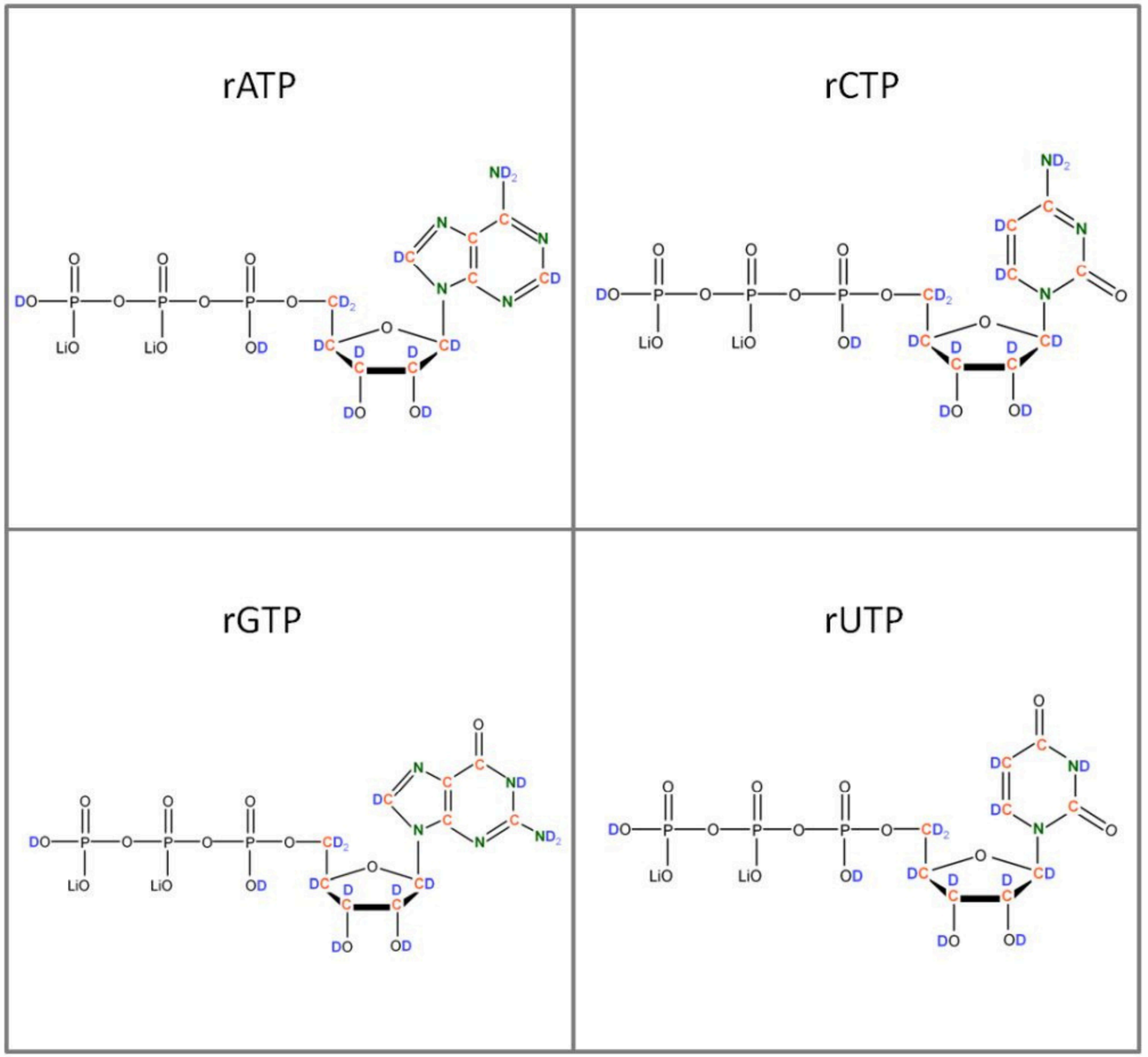
- Promoter: sequence that transcription apparatus recognizes and binds
- RNA coding region: DNA sequence that can be copied into RNA
- Terminator: sequence that signal where the transcription ends



Substrate of transcription: ribose nucleotides

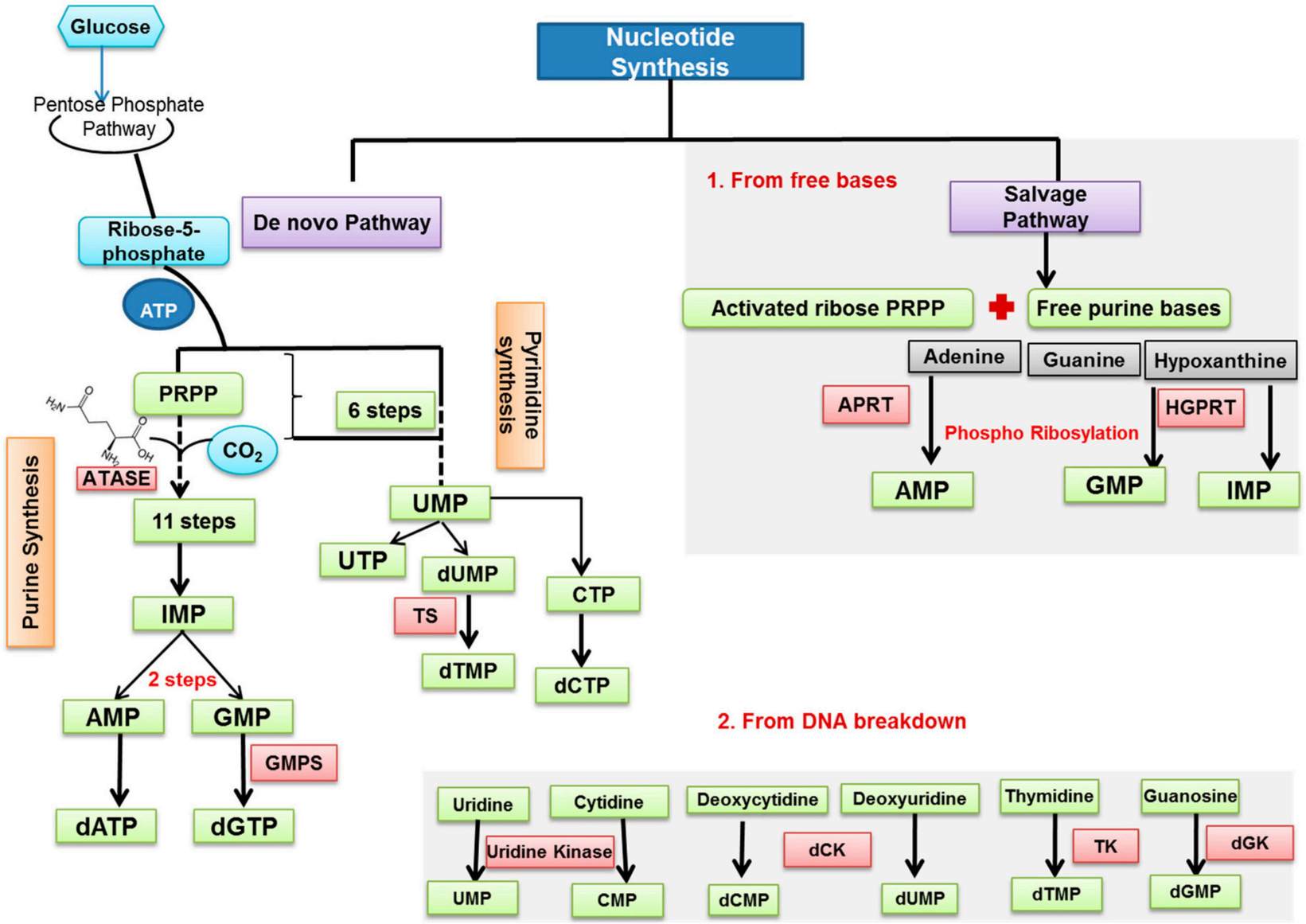


Where do these nucleotides come from ??



Nucleotides synthesis:

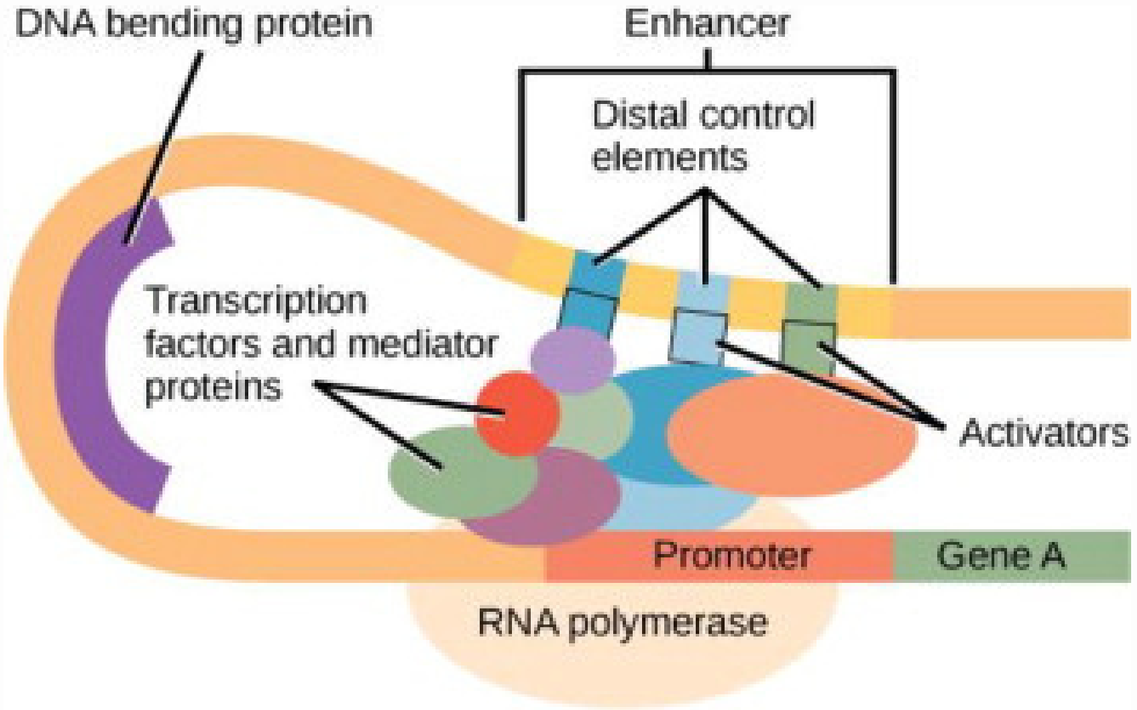
- De novo pathway
- Salvage pathway



Transcription apparatus: similar to DNA synthesis, this requires number of protein catalysts and RNA polymerase

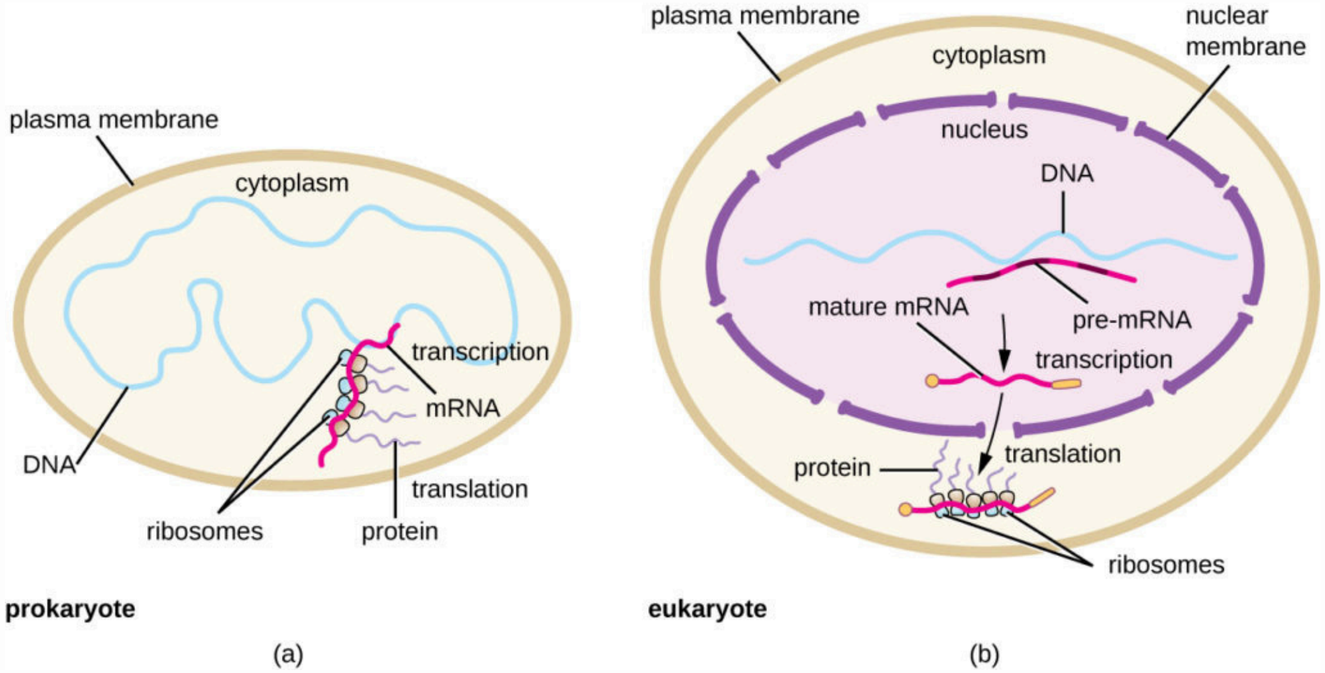
Transcription Key Components

- RNA polymerase, DNA template (gene + promoter + terminator), rNTPs, transcription factors, Mg^{2+}, accessory factors.
- Contrast with DNA polymerase: RNA Poly. needs **no primer**, unwinds DNA itself, and has lower fidelity (~1 error per 10^4–10^5 nt vs ~10^9 for replication) — acceptable because RNAs are short-lived copies.
- Energy: each incoming NTP releases pyrophosphate (PPi); PPi hydrolysis drives the reaction forward.



Where it happens ???

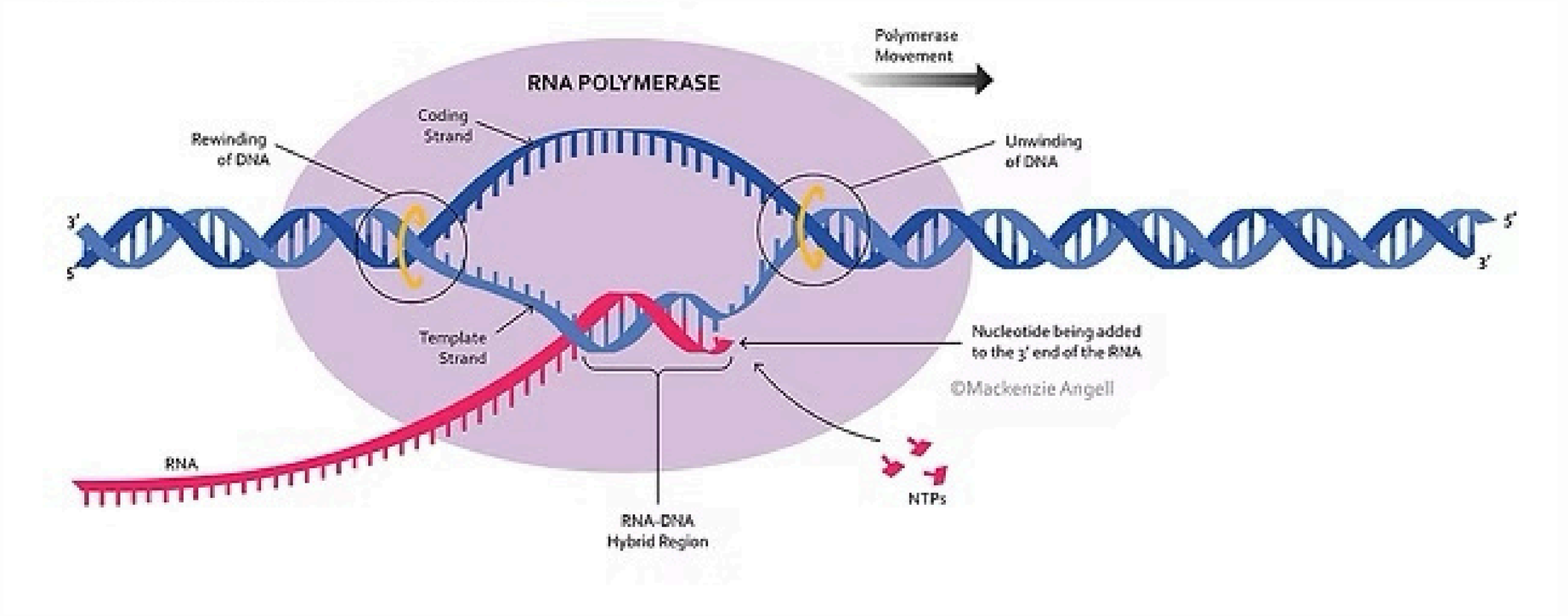
- **Bacteria: cytoplasm**, with no nuclear envelope, so ribosomes can start translating mRNA while it is still being made (coupled transcription–translation).
- **Eukaryotes: nucleus**, plus **mitochondria** and **chloroplasts** (own genomes). The nuclear envelope separates transcription from translation, which allows RNA processing before export.



Source: *Biology for Majors I, Module 10: DNA Transcription and Translation*

Template Strand and Coding Strand in Transcription

- RNA Poly. reads the template (antisense) strand 3′→5′ and builds RNA 5′→3′.
- The RNA matches the coding (sense) strand, with U instead of T. Sequences are written by convention as the coding strand.
- Which strand is the template differs between genes; neighbouring genes can be on opposite strands.



Template strand

3′ — TACGGTA — 5′

Coding strand

RNA transcript

Template strand

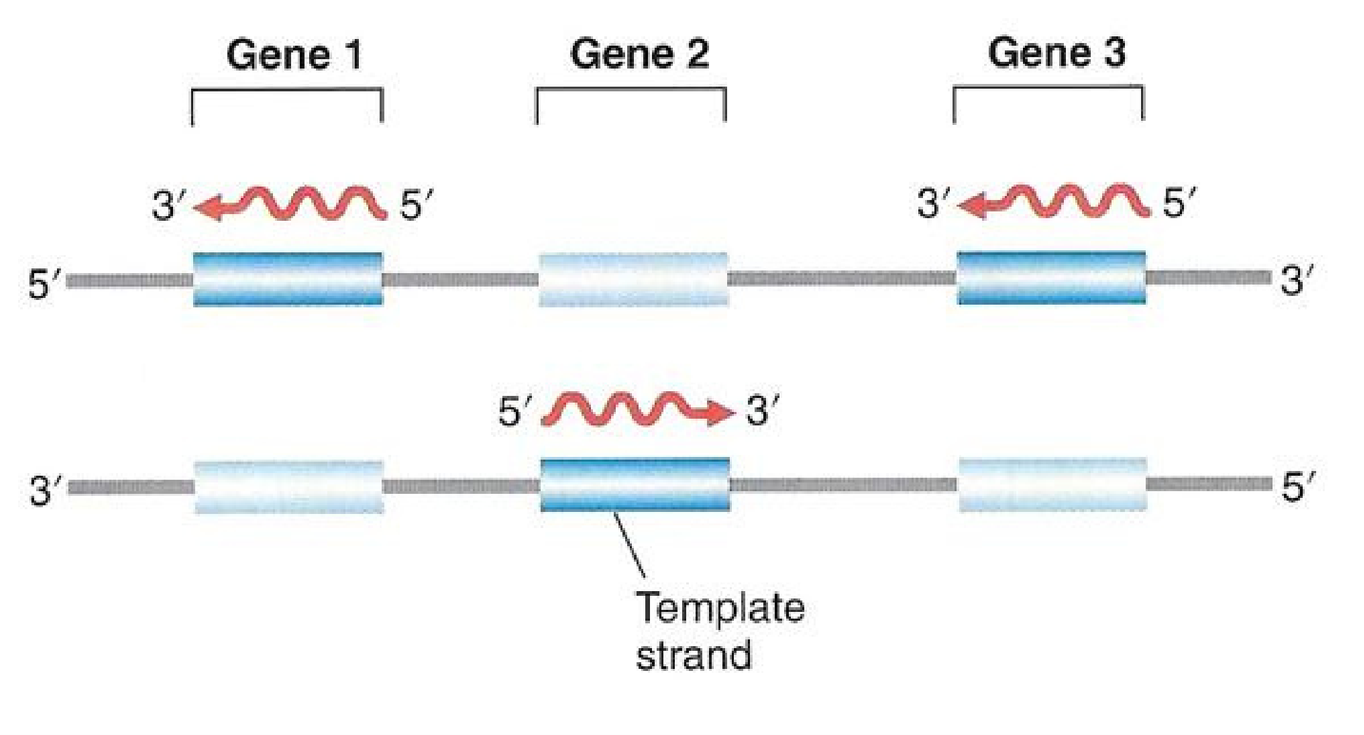
3′ — TACGGTA — 5′

Coding strand

5′ — ATGCCAT — 3′

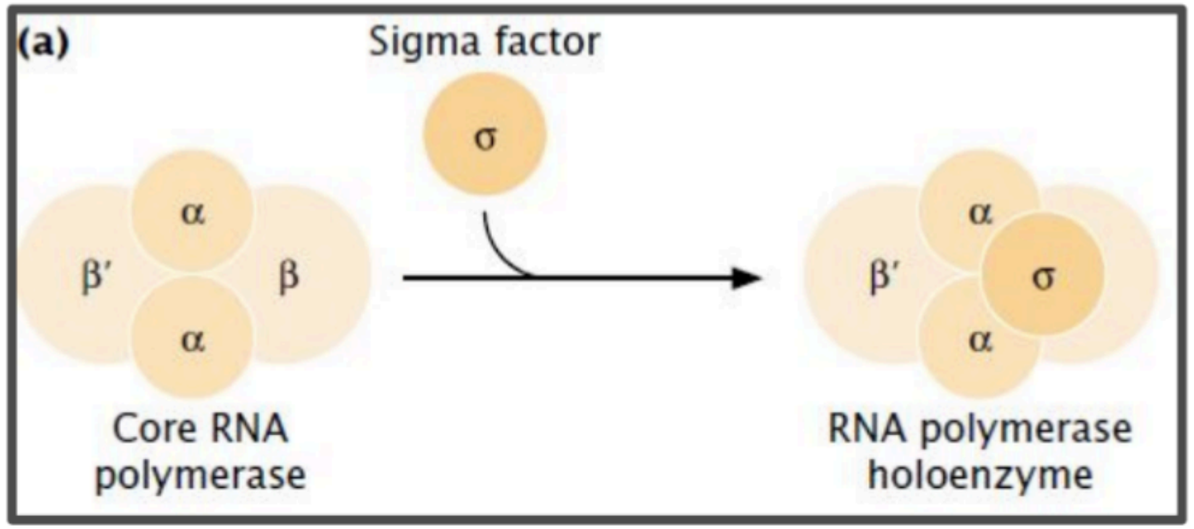
RNA transcript

5′ — AUGCCAU — 3′



Bacterial RNA polymerase:

- Normally possesses of one RNA polymerase which transcribe all types of RNA (mRNA, tRNA, rRNA)
- Large and multimeric enzyme: contain 5 protein subunits make the core enzyme work to elongate the RNA molecules $\alpha 2\beta\beta'\omega$
- The sigma factor controls the binding of RNA pol. to the promoter, different types of sigma factor bond to different promoter sequence
- Rho and NusA facilitate the termination of transcription



Eukaryotic RNA polymerase: eukaryotic cells possess 3 types of RNA polymerases I, II, III

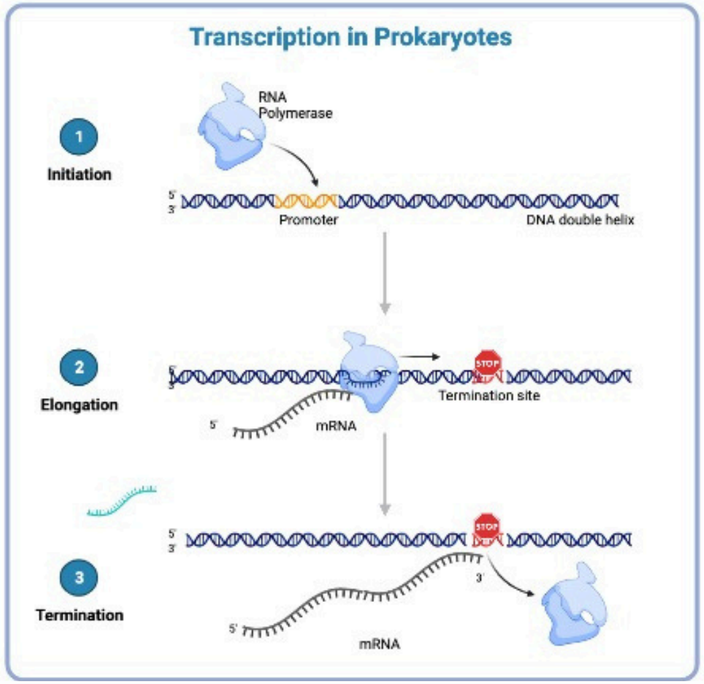
- Pol I: transcribe rRNA
- Pol II: transcribe pre-mRNA
- Pol III: transcribe small RNA molecules, tRNA, small rRNA and some snRNA
- All polymerases are large, contaning a dozen of protein subunits, some subunits are common to all polymerases, some are specific to come types
- Contains number of accessory proteins bind to core protein and affect its functions

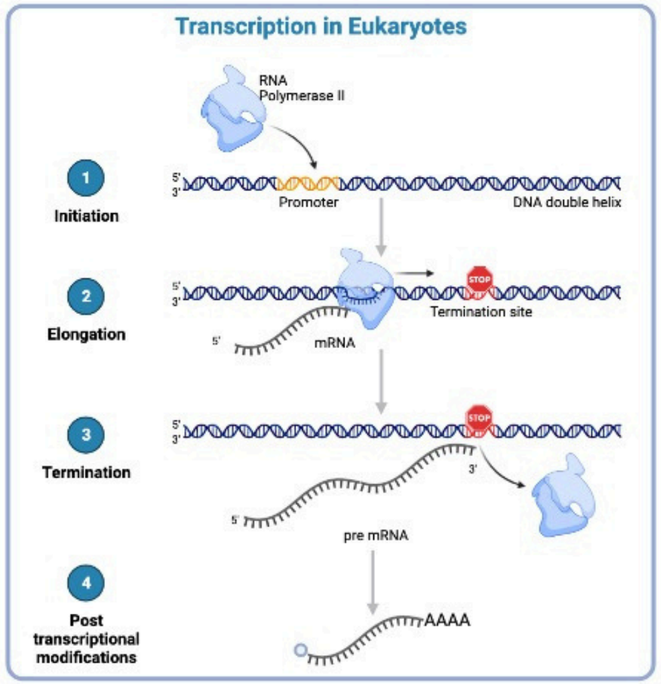
Table 13.3 Eukaryotic RNA polymerases

Type	Transcribes
RNA polymerase I	Large rRNAs
RNA polymerase II	Pre-mRNA, some snRNAs, snoRNAs
RNA polymerase III	tRNAs, small rRNA, snRNAs

Process of Transcription

1. Initiation, in which the transcription apparatus assembles on the promoter and begins the synthesis of RNA;
2. Elongation, in which RNA polymerase moves along the DNA, unwinding it and adding new nucleotides, one at a time, to the 3' end of the growing RNA strand; and
3. Termination, the recognition of the end of the transcription unit and the separation of the RNA molecule from the DNA template.





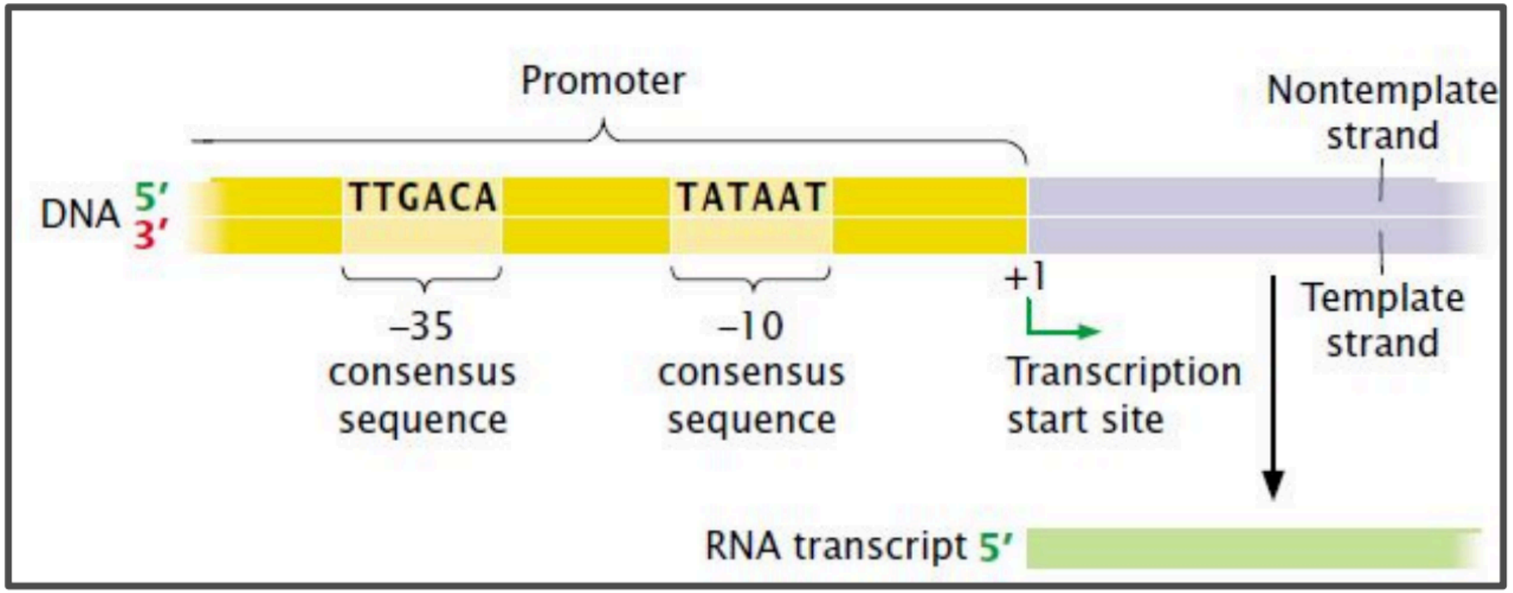
Gene Structure

- Prokaryotic genes: promoter, often operons (several genes, one polycistronic mRNA), no introns.
- Eukaryotic genes: core promoter + distant enhancers/silencers, exons and introns, one gene → one monocistronic mRNA.

Bacterial Transcription

Initiation:

- Promoter
- Formation of the transcription bubble
- Creation of the first bonds between rNTPs
- Escape of the transcription apparatus of the promoter

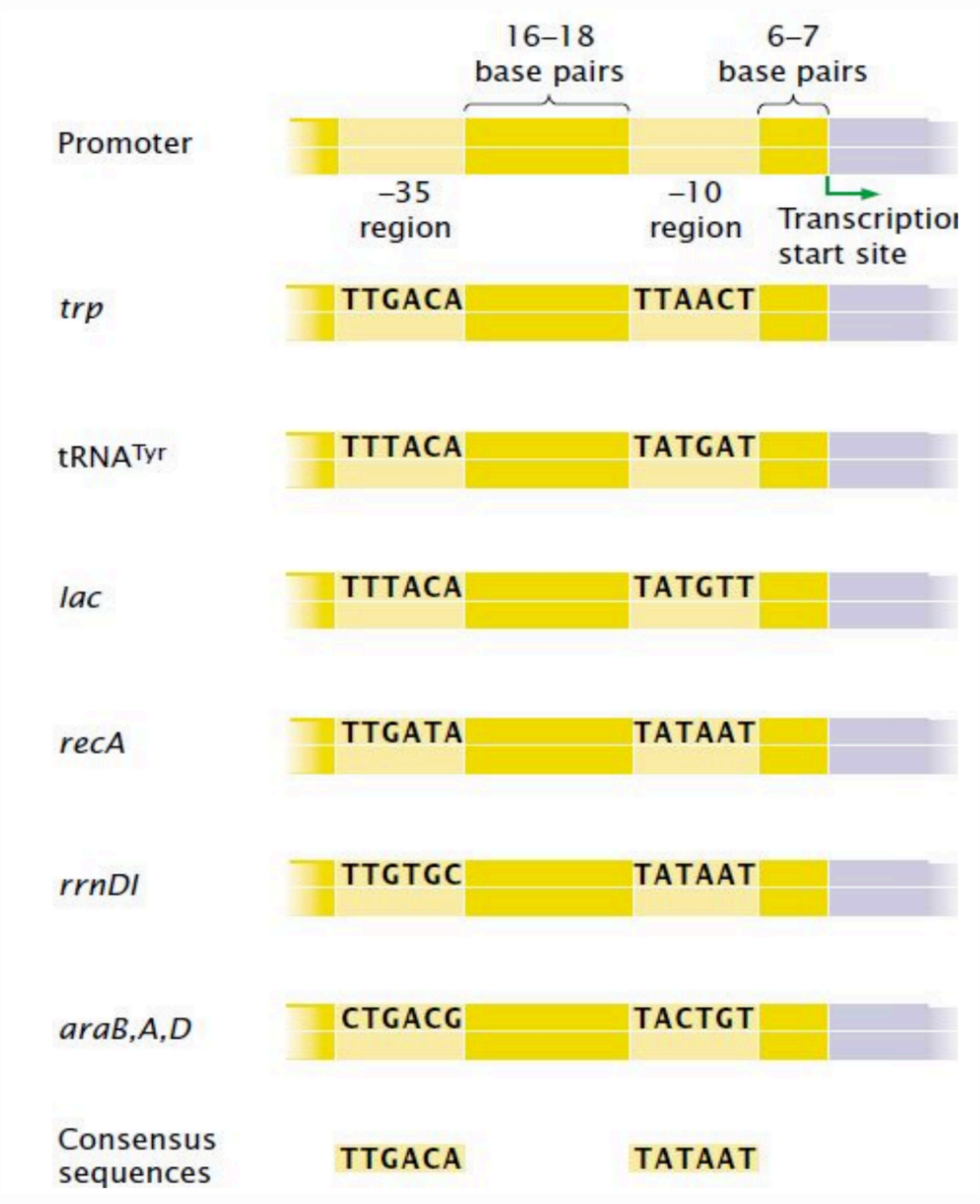


Bacterial promoter:

- Select which strand to read
- Direct the transcription
- Contains the consensus sequences

$$\text{Consensus sequence} = \frac{\begin{array}{r} 5'-\text{A A T A A A}-3' \\ 5'-\text{T T T A A T}-3' \\ 5'-\text{T A T T T T}-3' \\ 5'-\text{T A A A A T}-3' \end{array}}{5'-\text{T A T A A T}-3'}$$

- Lies at app. within -10 and -35 of the transcription unit, it is usually TAGACA


16–18
base pairs
6–7
base pairs
Promoter
–35
region
–10
region
Transcription
start site
trp
TTGACA
TTAACT
tRNATyr
TTTACA
TATGAT
lac
TTTACA
TATGTT
recA
TTGATA
TATAAT
rrnDI
TTGTGC
TATAAT
araB,A,D
CTGACG
TACTGT
Consensus
sequences
TTGACA
TATAAT

Exercise: find the consensus of the −35 box

These are the −35 regions of six E. coli promoters, lined up and written on the coding strand (5′→3′):

Promoter 1: T T G A C A

Promoter 2: T T G A T A

Promoter 3: C T G A C A

Promoter 4: T T T A C A

Promoter 5: T T G A C T

Promoter 6: T A G A C A

Position: 1 2 3 4 5 6

a) Work out the consensus sequence.

b) A new promoter has the −35 region TAGATT. How many positions match the consensus, and would you expect it to be a strong or weak promoter?

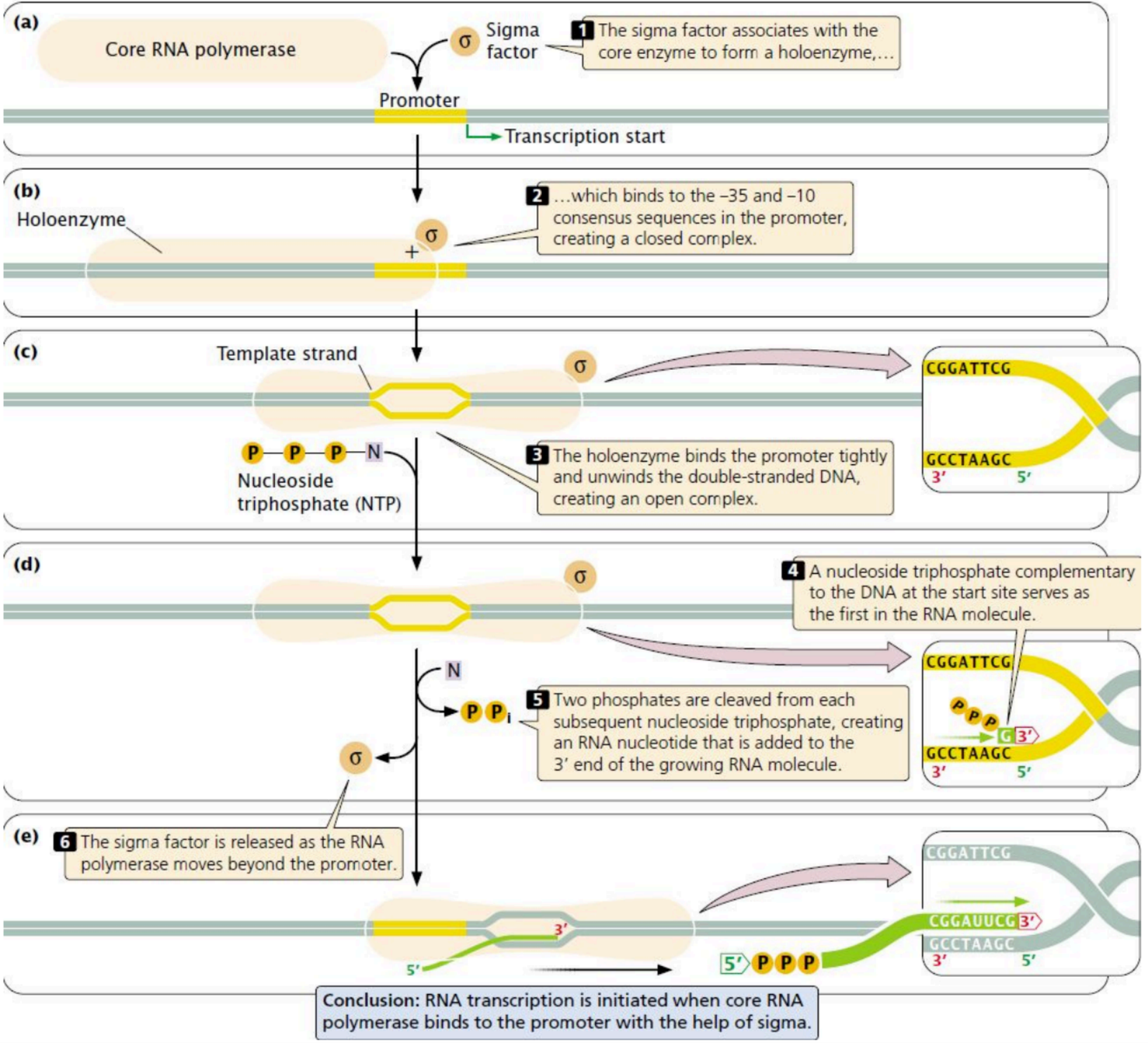
Promoter 1: T T G A C A
Promoter 2: T T G A T A
Promoter 3: C T G A C A
Promoter 4: T T T A C A
Promoter 5: T T G A C T
Promoter 6: T A G A C A
Position: 1 2 3 4 5 6

a) The consensus is 5′-TTGACA-3′. Only Promoter 1 matches it exactly; each of the others differs at one position.

b) Compare TAGATT with TTGACA: T a G A t t. Positions 1, 3 and 4 match, so that's 3 out of 6. With that few matches, σ factor would bind poorly, so this is probably a weak promoter with low transcription.

Initiation: Four steps

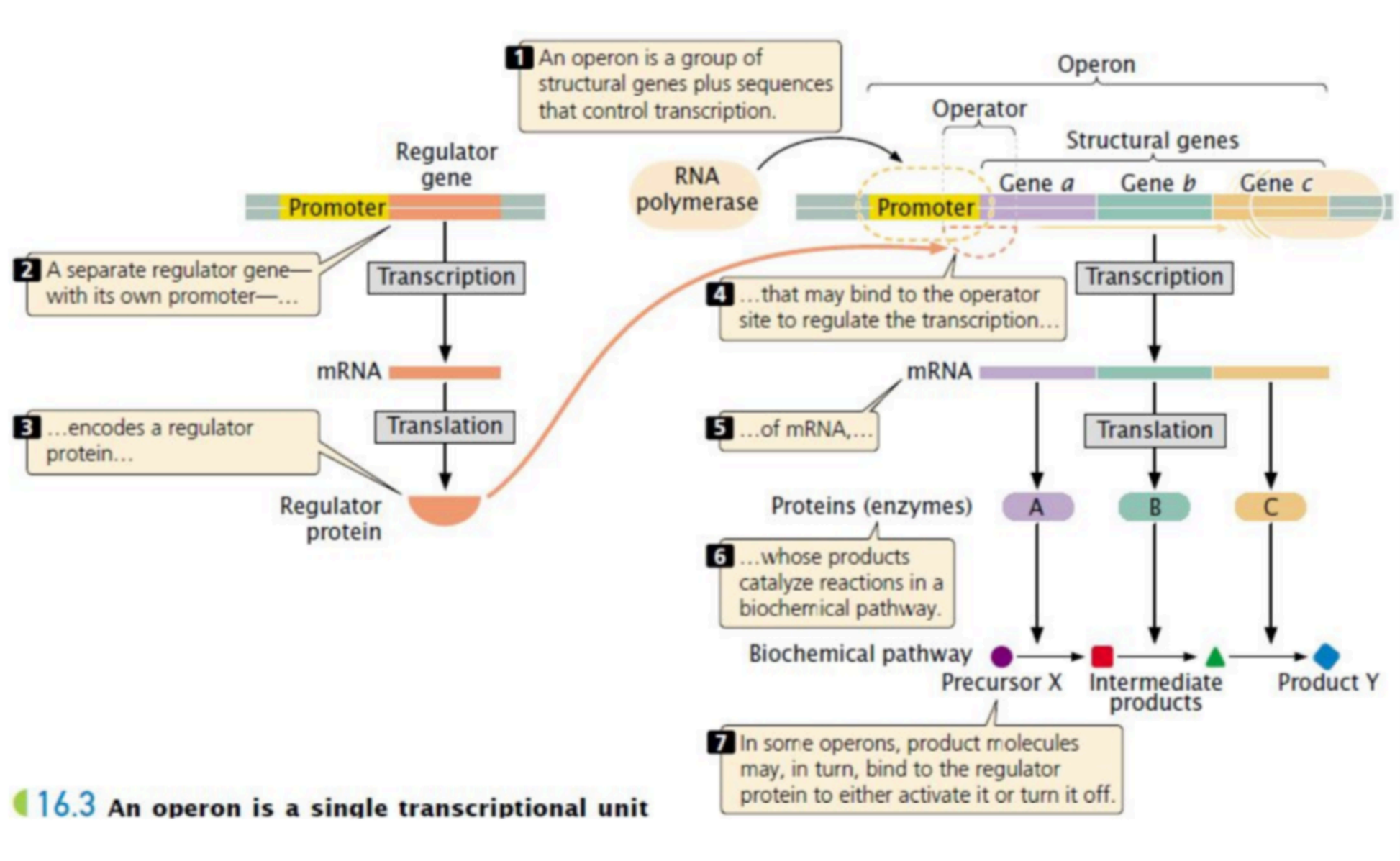
- Promoter recognition & binding (Closed complex): σ region 4 reads −35, region 2 reads −10; DNA still double-stranded.
- Open complex: core RNAP melts DNA near TSS (AT-rich −10 helps), forming the transcription bubble. No ATP needed.
- Abortive initiation: RNAP makes and releases short RNAs (2–9 nt) repeatedly while staying on the promoter. "Scrunching" — RNAP pulls downstream DNA into itself, storing strain.
- Promoter escape: once RNA reaches ~10 nt, the stored strain breaks promoter contacts; σ is usually released and recycled.


(a)
Core RNA polymerase
σ
Sigma factor
1 The sigma factor associates with the core enzyme to form a holoenzyme,…
Promoter
Transcription start
(b)
Holoenzyme
σ
+
2 …which binds to the –35 and –10 consensus sequences in the promoter, creating a closed complex.
(c)
Template strand
σ
CGGATTCG
GCCTAAGC
3′
5′
P P P N
Nucleoside triphosphate (NTP)
3 The holoenzyme binds the promoter tightly and unwinds the double-stranded DNA, creating an open complex.
(d)
σ
4 A nucleoside triphosphate complementary to the DNA at the start site serves as the first in the RNA molecule.
CGGATTCG
N
P P i
5 Two phosphates are cleaved from each subsequent nucleoside triphosphate, creating an RNA nucleotide that is added to the 3′ end of the growing RNA molecule.
G 3′
GCCTAAGC
3′
5′
σ
(e)
6 The sigma factor is released as the RNA polymerase moves beyond the promoter.
CGGATTCG
CGGAUUCG 3′
GCCTAAGC
3′
5′
3′
5′
5′ P P P
Conclusion: RNA transcription is initiated when core RNA polymerase binds to the promoter with the help of sigma.

Elongation

- Core RNAP moves ~40–80 nt/s, with a bubble of ~12–14 bp and an RNA–DNA hybrid of ~8–9 bp.
- Twin-domain model: positive supercoils build ahead, negative behind; DNA gyrase and topoisomerase I relieve them.
- Proofreading: RNAP can backtrack, and GreA/GreB stimulate cleavage of a misincorporated 3′ end.
- RNA polymerase acts as a stable linker between the DNA template and the nascent RNA strands to ensure that elongation is not interrupted prematurely.

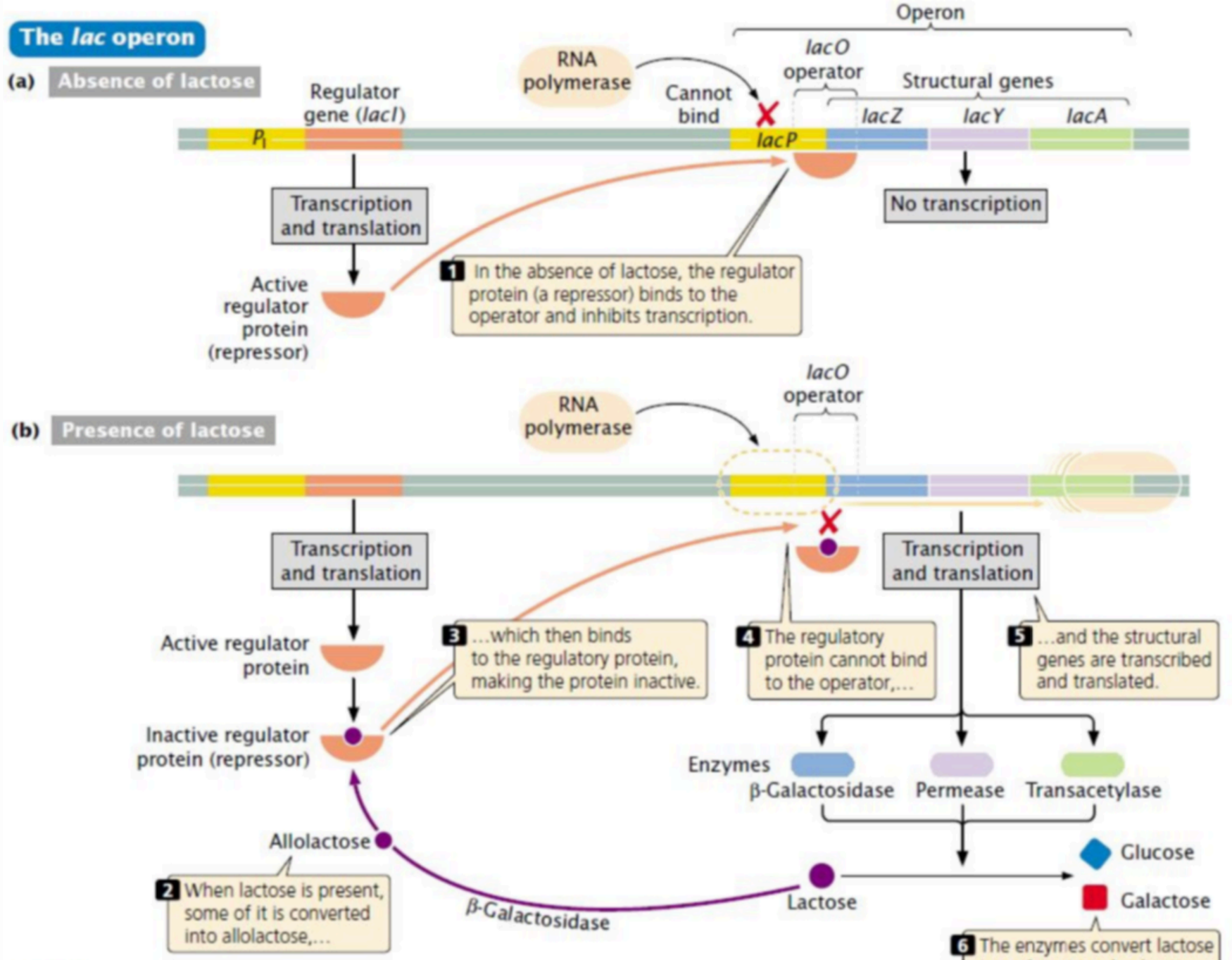
Transcription Regulation in Prokaryotes



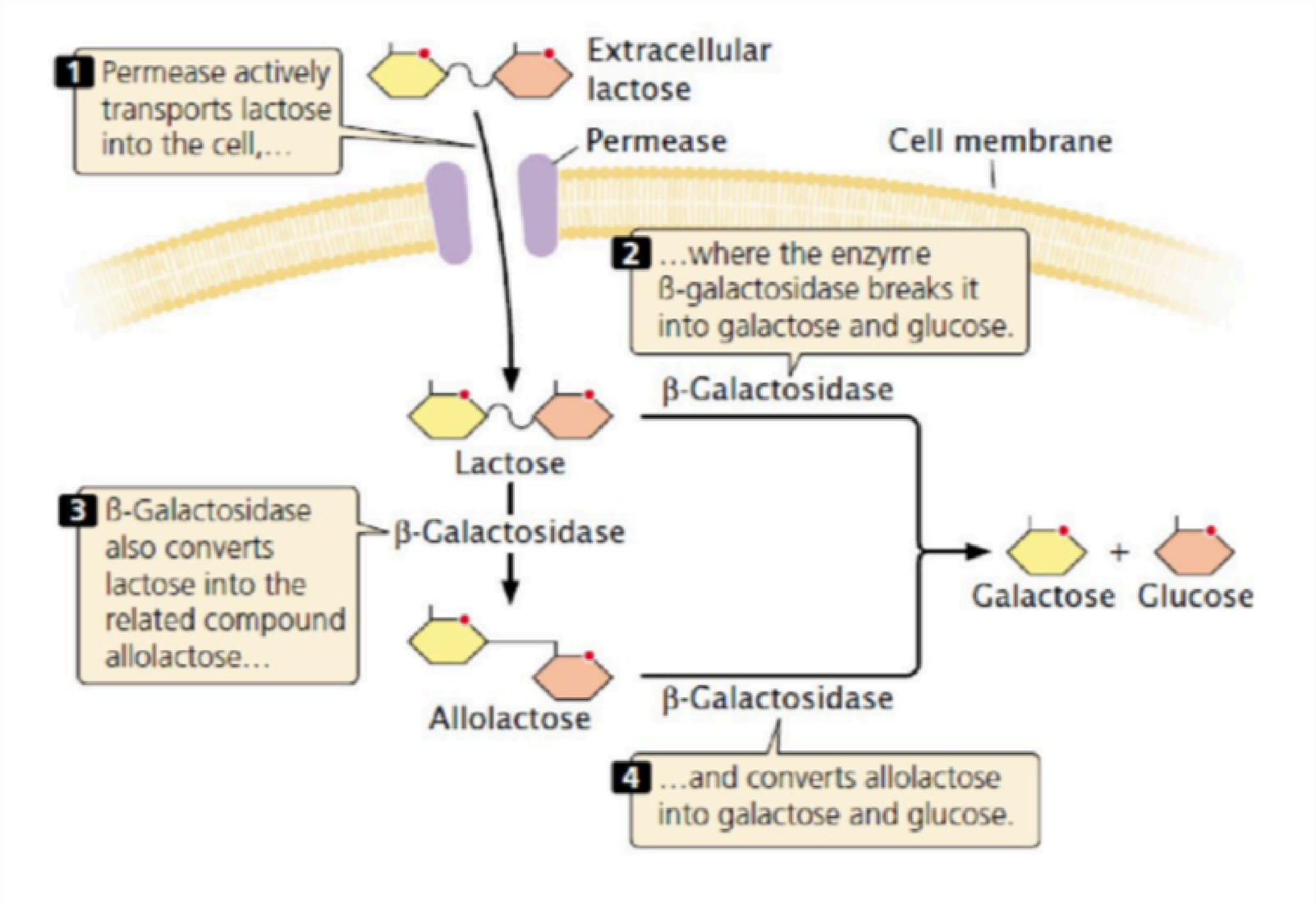
16.3 **An operon is a single transcriptional unit**

Negative and positive control: inducible and repressible operon

- negative control: a regulatory protein acts as a repressor, binding to DNA and inhibiting transcription;
- positive control: a regulatory protein acts as an activator, stimulating transcription.

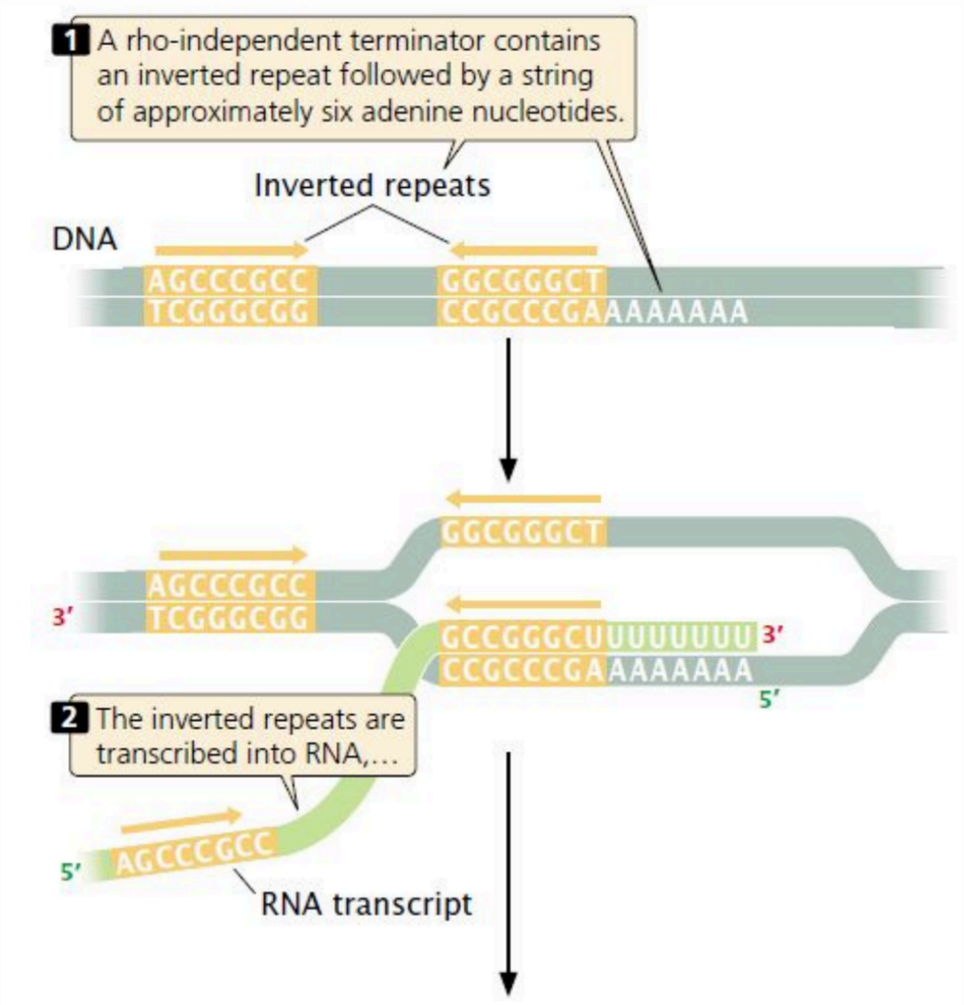


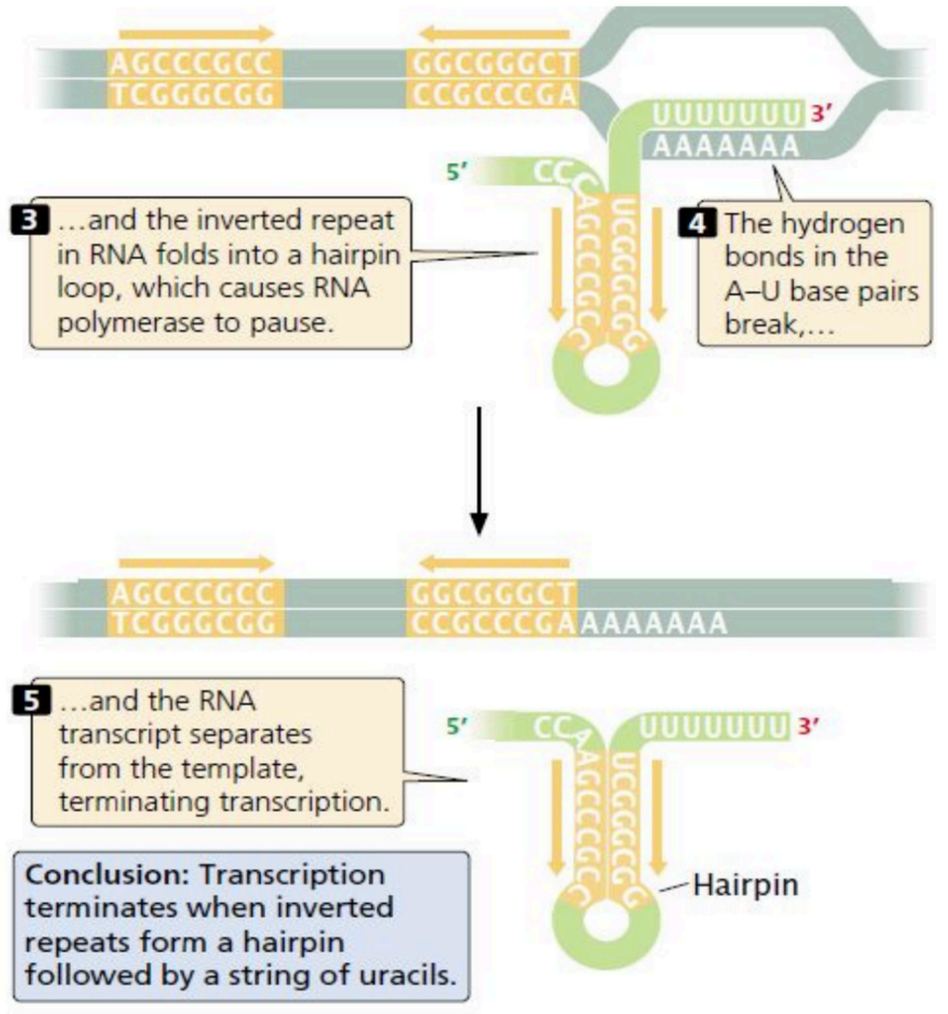
16.7 **The *lac* operon regulates lactose metabolism.**


1 Permease actively transports lactose into the cell,...
Extracellular lactose
Permease
Cell membrane
2 ...where the enzyme β-galactosidase breaks it into galactose and glucose.
β-Galactosidase
Lactose
3 β-Galactosidase also converts lactose into the related compound allolactose...
β-Galactosidase
Allolactose
β-Galactosidase
Galactose + Glucose
4 ...and converts allolactose into galactose and glucose.

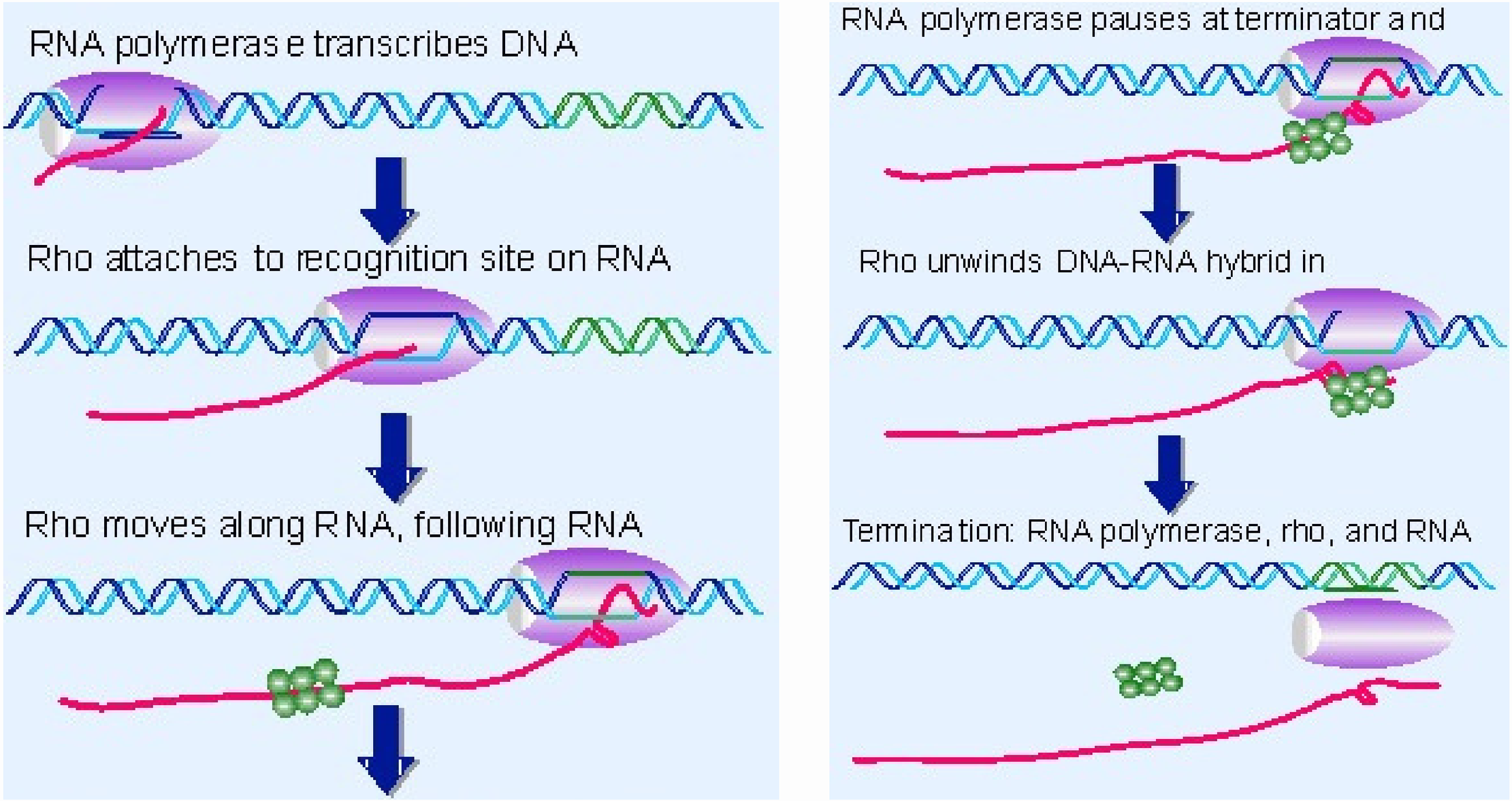
Termination

Rho-independent Termination

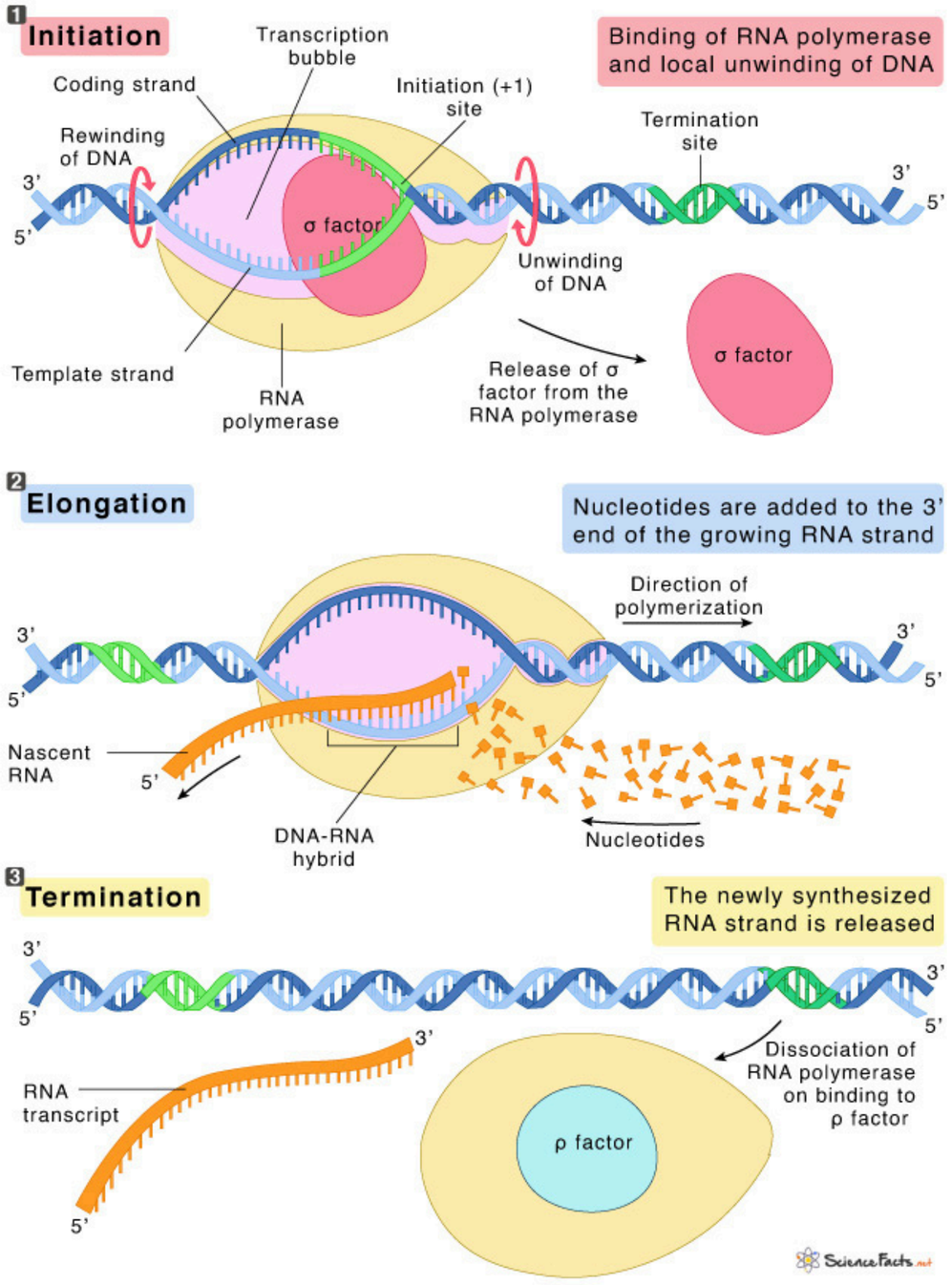




Rho-dependent Termination



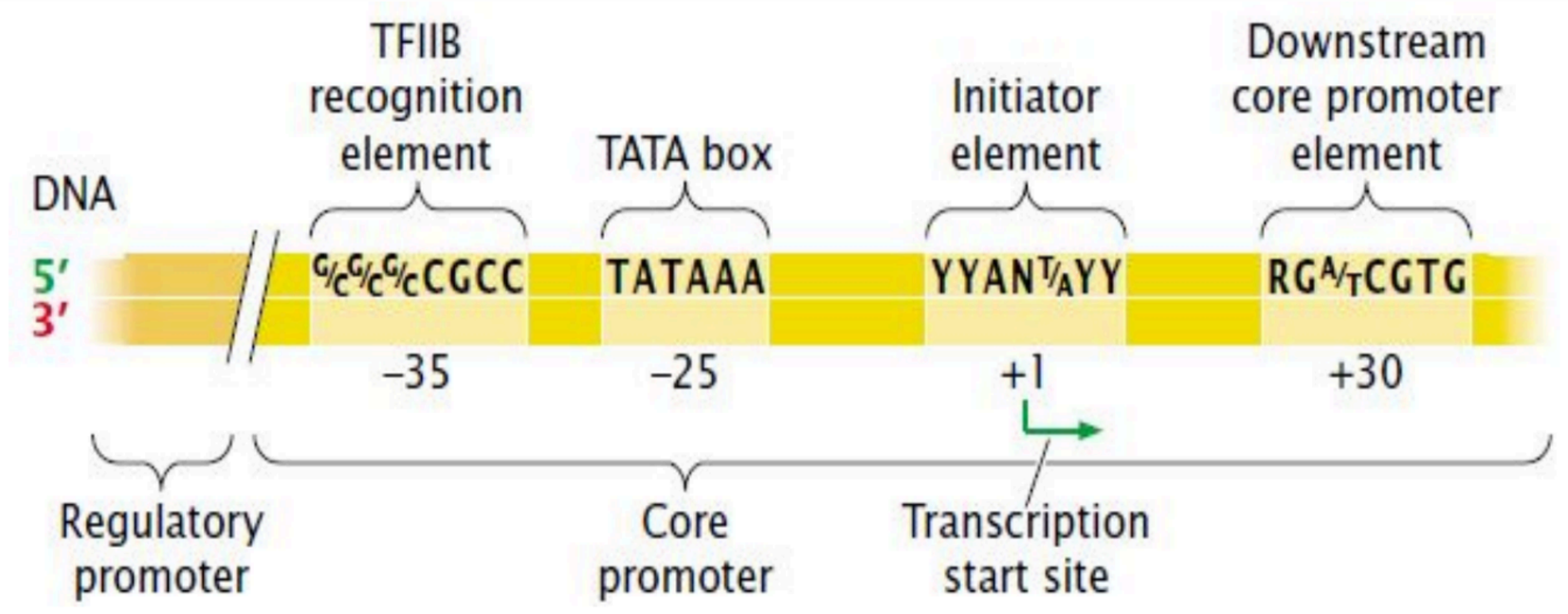
Steps of Transcription In Prokaryotes



Eukaryotic Transcription

- Has differnet polymerases involved → different promoter specified
- With each promoter type, specific protein included to bind RNA polymerase to DNA
- Chromosome is modified before transcription occurs (acetyltransferase, chromatin remodelling proteins..)
- Require the transcriptional activator proteins

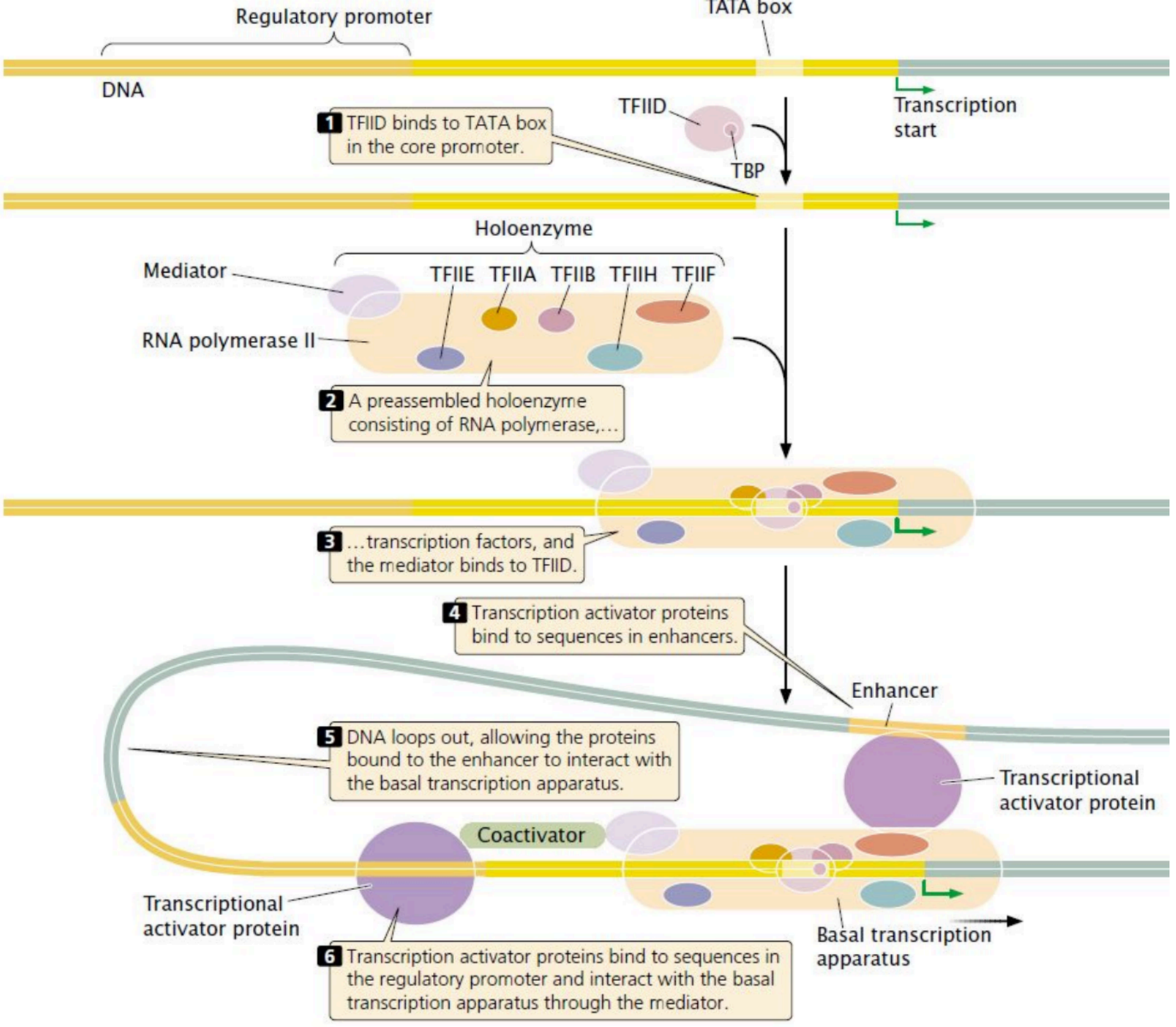
The RNA Polymerase II promoter



Initiation: building the pre-initiation complex (PIC)

1. TFIID (TBP + TAFs) binds the core promoter. TBP sits in the minor groove and bends DNA ~80°; TAFs read Inr/DPE at TATA-less promoters.
2. TFIIA stabilises TBP–DNA; TFIIB binds next to TBP and sets the start site.
3. Pol II + TFIIF join.
4. TFIIE recruits TFIIH.
5. TFIIH does two jobs: its helicase (XPB) opens the DNA (ATP-dependent, unlike bacteria), and its kinase (CDK7) phosphorylates CTD Ser5 → promoter escape.

- Memory aid: "D-A-B-F-E-H" order of assembly.
- TFIIH also works in nucleotide excision repair.


Regulatory promoter
TATA box
DNA
Transcription start
TFIID
TBP
1 TFIID binds to TATA box in the core promoter.
Holoenzyme
Mediator
TFIIE TFIIA TFIIB TFIIH TFIIF
RNA polymerase II
2 A preassembled holoenzyme consisting of RNA polymerase,…
3 …transcription factors, and the mediator binds to TFIID.
4 Transcription activator proteins bind to sequences in enhancers.
Enhancer
5 DNA loops out, allowing the proteins bound to the enhancer to interact with the basal transcription apparatus.
Transcriptional activator protein
Coactivator
Transcriptional activator protein
Basal transcription apparatus
6 Transcription activator proteins bind to sequences in the regulatory promoter and interact with the basal transcription apparatus through the mediator.

Elongation

- After ~20–60 nt, many metazoan genes show promoter-proximal pausing, held by DSIF and NELF.
- P-TEFb (CDK9) phosphorylates CTD Ser2, DSIF and NELF → productive elongation.
- CTD code: Ser5-P near the 5′ end (capping enzymes), Ser2-P rising toward the 3′ end (splicing and 3′-processing factors). This couples transcription to processing.
- Chromatin: FACT and Spt6 remove and replace nucleosomes as Pol II passes.

Regulation

- Activators bind enhancers; DNA loops bring them to the promoter via Mediator and co-activators (e.g. histone acetyltransferases).
- Silencers bind repressors that recruit co-repressors (e.g. histone deacetylases).

Termination:

- is less understood in eukaryote than in prokaryote
- Different polymerases use different mechanisms to terminate transcription
- Termination may end at position that far from the stop site
- May couple with cleavage

Termination:

- No defined terminator. Pol II transcribes the PAS (AAUAAA); CPSF binds it, CstF binds a GU-rich element downstream.
- The RNA is cleaved ~10–30 nt downstream of AAUAAA (often after CA); poly(A) polymerase then adds the tail.
- Torpedo model: Xrn2 (Rat1 in yeast) degrades the uncapped downstream RNA 5′→3′ and dislodges Pol II.
- Allosteric model: passing the PAS changes Pol II's factors so it becomes termination-prone. Current view: both act together.
- Pol I and Pol III terminate differently (Pol III at a run of T's, a nice echo of bacterial intrinsic terminators).

What happens if AAUAAA mutates into AAGAAA?

Practice - Fill in the blank

1. The coding strand reads 5′-ATGCCAGTA-3′. The mRNA is 5′-_____-3′ and the template strand is 3′-_____-5′.
2. RNA polymerase synthesizes RNA in the _____ direction while reading the template strand _____.
3. The E. coli core RNA polymerase has the subunit composition _____; adding the _____ factor forms the holoenzyme.
4. In eukaryotic transcription, RNA polymerase II transcribes _____ and most snRNAs; RNA polymerase III transcribes _____ and 5S rRNA.
5. _____ binds next to TBP and helps choose the transcription start site.
6. _____ are regulatory DNA elements that can increase transcription from kilobases away through DNA _____.

Practice - Fill in the blank

1. The coding strand reads 5′-ATGCCAGTA-3′. The mRNA is **5′-AUGCCAGUA-3′** and the template strand is **3′-TACGGTCAT-5′**.
2. RNA polymerase synthesizes RNA in the **5′→3′** direction while reading the template strand **3′→5′**.
3. The E. coli core RNA polymerase has the subunit composition **$\alpha_2\beta\beta'\omega$**; adding the **σ (sigma)** factor forms the holoenzyme.
4. In eukaryotic transcription, RNA polymerase II transcribes **mRNA** and most snRNAs; RNA polymerase III transcribes **tRNA** and 5S rRNA.
5. **TFIIB** binds next to TBP and helps choose the transcription start site.
6. **Enhancers** are regulatory DNA elements that can increase transcription from kilobases away through DNA **looping.**

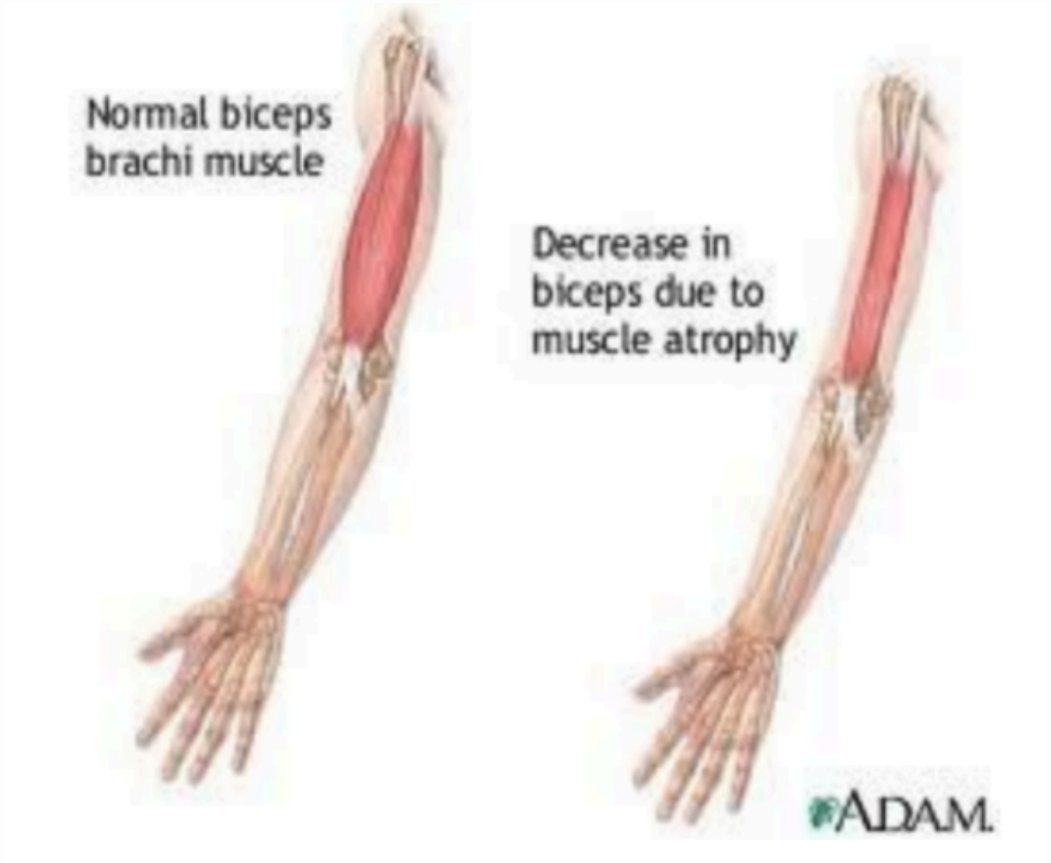
Practice - Fill in the table: Transcription in Prokaryotes and Eukaryotes

Feature	Prokaryotes	Eukaryotes
Location	cytoplasm	nucleus
Template	DNA 3' - 5'	DNA 3' - 5'
RNA Polymerase	core RNA Poly + sigma factor	RNA Poly I (large rRNA), II (mRNA), III (tRNA, small rRNA)
Gene Organisation	polycistronic	monocistronic
Introns	Absent	Presence
Coupling with translation	happen together	separate
Promoter recognition	sigma factor	transcription factor TFIID (TBP)
DNA opening	core RNA Poly + sigma factor, no ATP	TF, need ATP
Termination signal	rho dependent (has rho protein) and rho independent (inverted repeat)	poly(A) signal AAUAAA

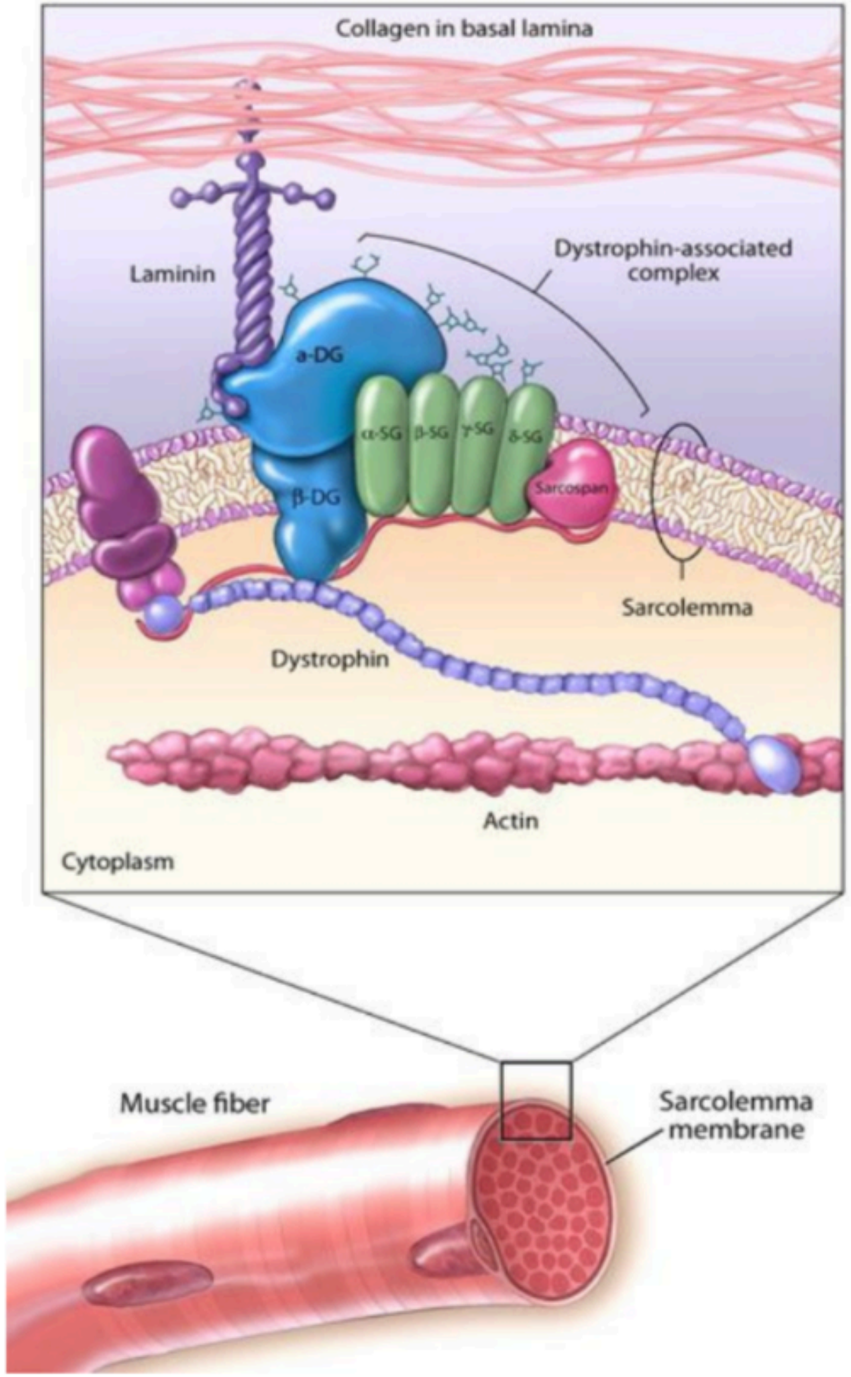
RNA Processing

The example of dystrophin gene and function - case study

Lack of dystrophin causes muscle biceps decrease → atrophy → problem in movement, breathing... (Dunchenne muscular dystrophy)



https://www.youtube.com/watch?v=Ebu8W8Osuxk&t=125s



The dystrophin

- The intracellular cytoskeleton protein
- Anchor the intramuscular filament to plasma membrane
- Lies just under plasma membrane

The dystrophin gene

- Immense, consist of more than 2 million nucleotides
- Only 12,000 nucleotides encode amino acids
- On X chromosome

- Dystrophin gene should be modified to be functional
- Where/How to be modified ???

Concept of gene

- Before: gene is inherited factor that determine trait
- Later: gene is set of nucleotides that specify the product (RNA nucleotide sequence or amino acid sequence)

About 99% sequence of dystrophin gene is not coded for amino acids

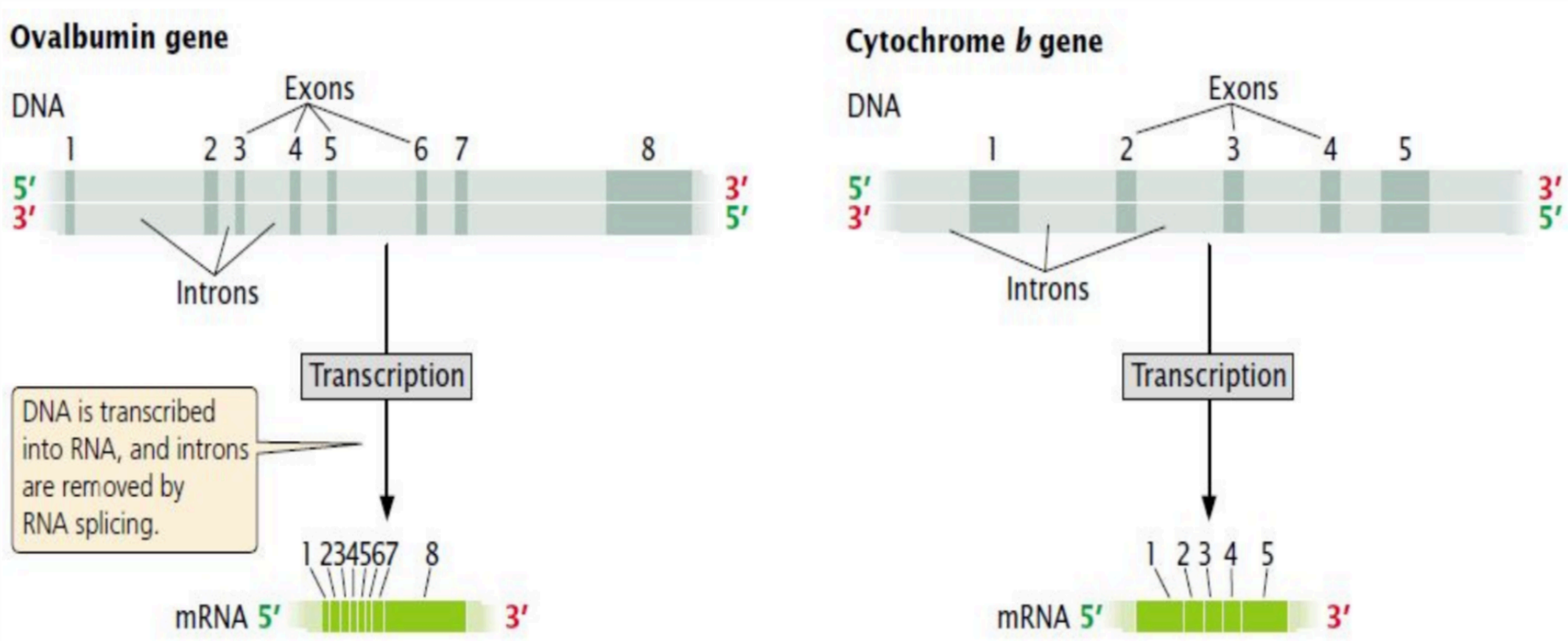
Gene Structure

Gene is organized differently in prokaryote and eukaryote

- In prokaryote: gene is organized collinearly with protein, there is correspondence between nucleotide sequence and amino acid sequence
- In eukaryote: the nucleotide sequence is much longer than amino acid sequence, large sequence in eukaryotic genes does not code for amino acid

Intron

- is the sequence in eukaryotic gene that is not translated into amino acids
- Size and number of intron is directly related to the organismal complexity



There are 4 major types of introns

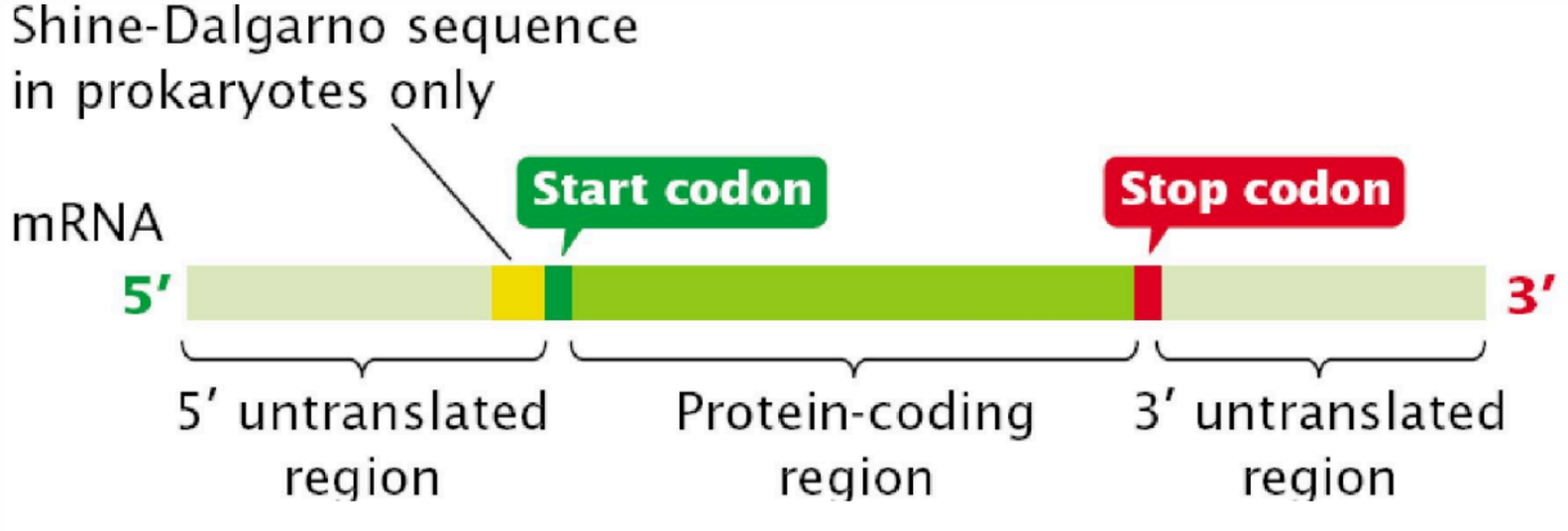
Table 14.1 Major types of introns

Type of Intron	Location	Type of Splicing
Group I	Some rRNA genes	Self-splicing
Group II	Protein-encoding genes in mitochondria and chloroplasts	Self-splicing
Nuclear pre-mRNA	Protein-encoding genes in the nucleus	Spliceosomal
tRNA	tRNA genes	Enzymatic

- There are also several types of minor introns: group III introns, twintrons, archaeal introns

1. Changes in Messenger RNA

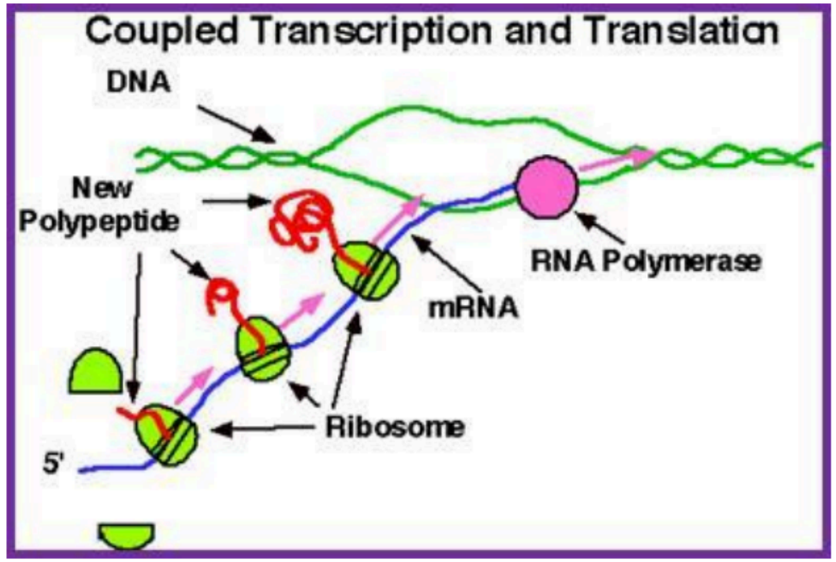
- Messenger RNA functions as a template for protein synthesis, it carries the genetic information from DNA to ribosome and help to assemble amino acids in their correct order
- Structure of mRNA



Pre-mRNA Processing

In prokaryote:

- Transcription and translation take place simultaneously → no opportunity to modify RNA



In eukaryote: pre-mRNA is modified before translation

- Addition of 5’ cap
- Addition of poly A tail
- Modification of the coding region

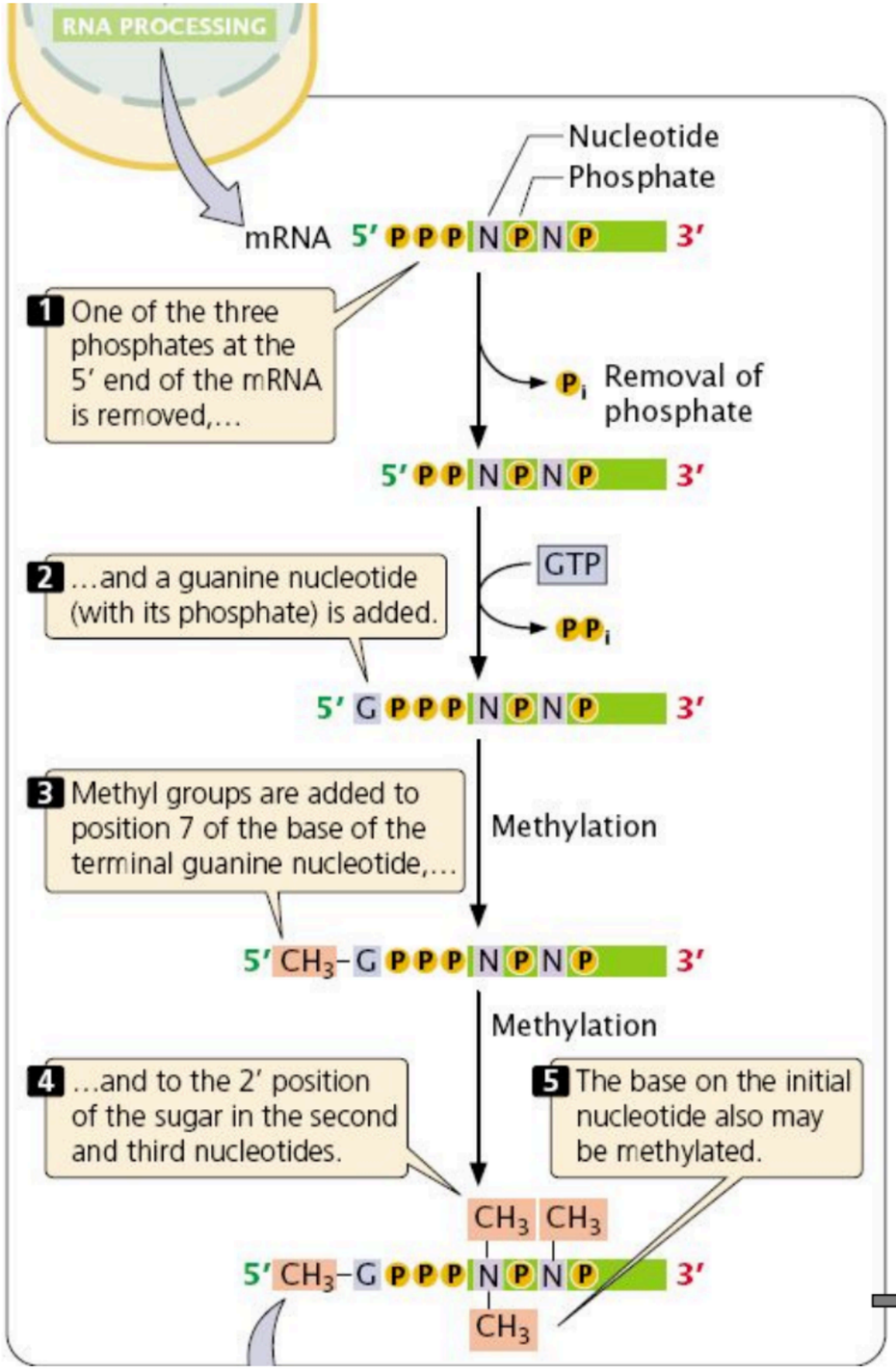
Addition of 5' cap

Cap prepares for the binding of ribosome and initiate the translation

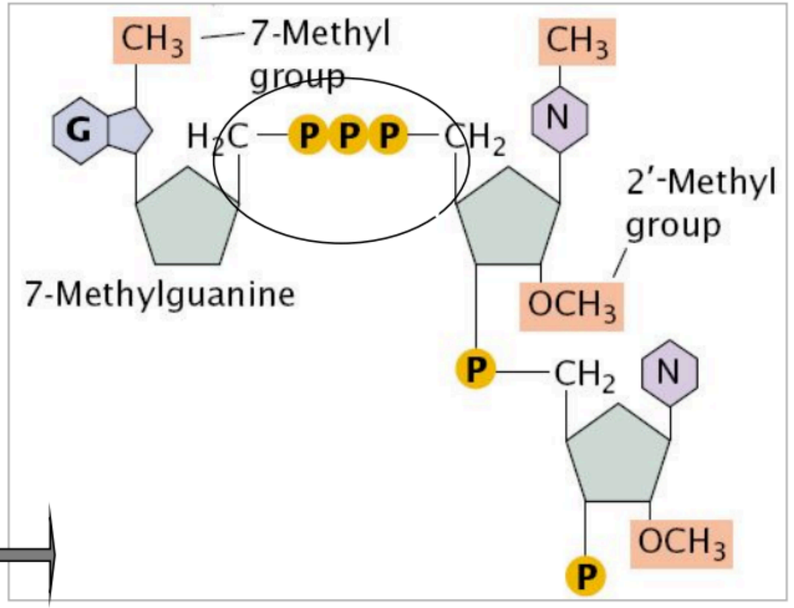
- Increase the stability of the mRNA
- Influence the splicing of introns

Cap formed by

- Addition of the extra nucleotides at the 5’ end of the mRNA
- Add methyl groups to the newly synthesized nucleotides

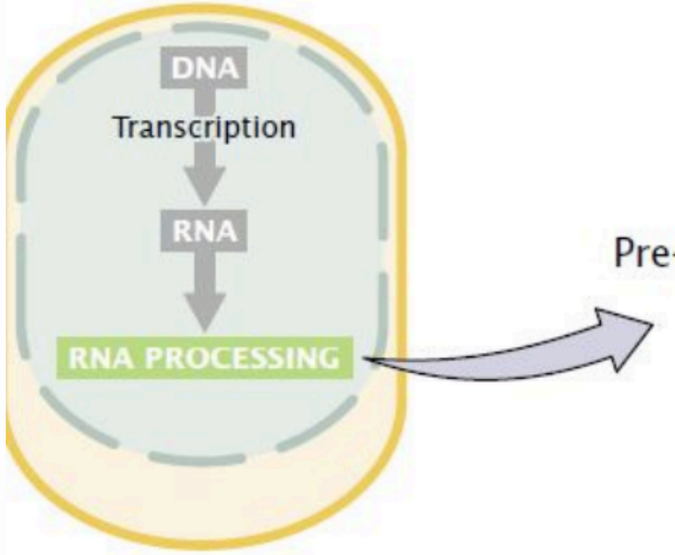


- Cap appears in all pre-mRNA
- one phosphate (P) group at the first nucleotide (N) is removed
- G is added shortly after the initiation of transcription and linked with the P group of the first N through the unique 5-5 linkage
- G and other N can be methylated (Me)
- Cap can be represented as 5’ (Me)G pppNpNpN…
- Methyl groups can be added to different positions (C7, N1, O2)



Addition of the polyA tail

- Most mature eukaryotic mRNAs have from 50-70 nucleotides A at the 3' end
- Polynucleotides A are not encoded in DNA but added to the mRNA after transcription by the process polyadenylation
- Pre-mRNA is proceeded by cutting the extra material at the 3’ end at a cleavage site and polyA is added



DNA

Transcription start site

Transcription

Consensus sequence

11–30 nucleotides

Pre-mRNA is cleaved, at a position from 11 to 30 nucleotides downstream of an AAUAAA consensus sequence, in the 3′ untranslated region.

Pre-mRNA 5′ AAUAAA 3′

Cleavage site

Cleavage

5′ AAUAAA 3′

The addition of adenine nucleotides (polyadenylation) takes place at the 3′ end of the pre-mRNA, generating the poly(A) tail.

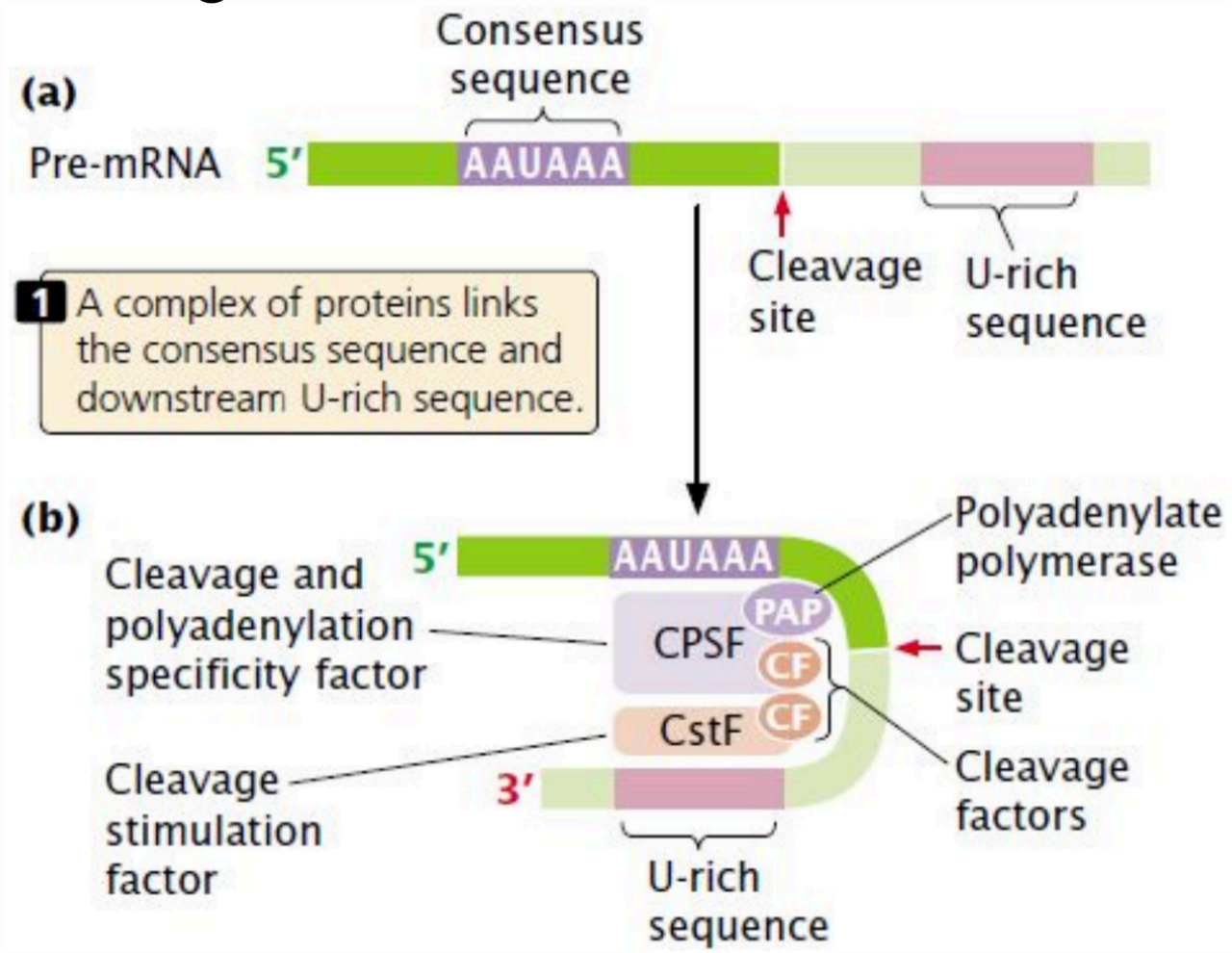
Polyadenylation

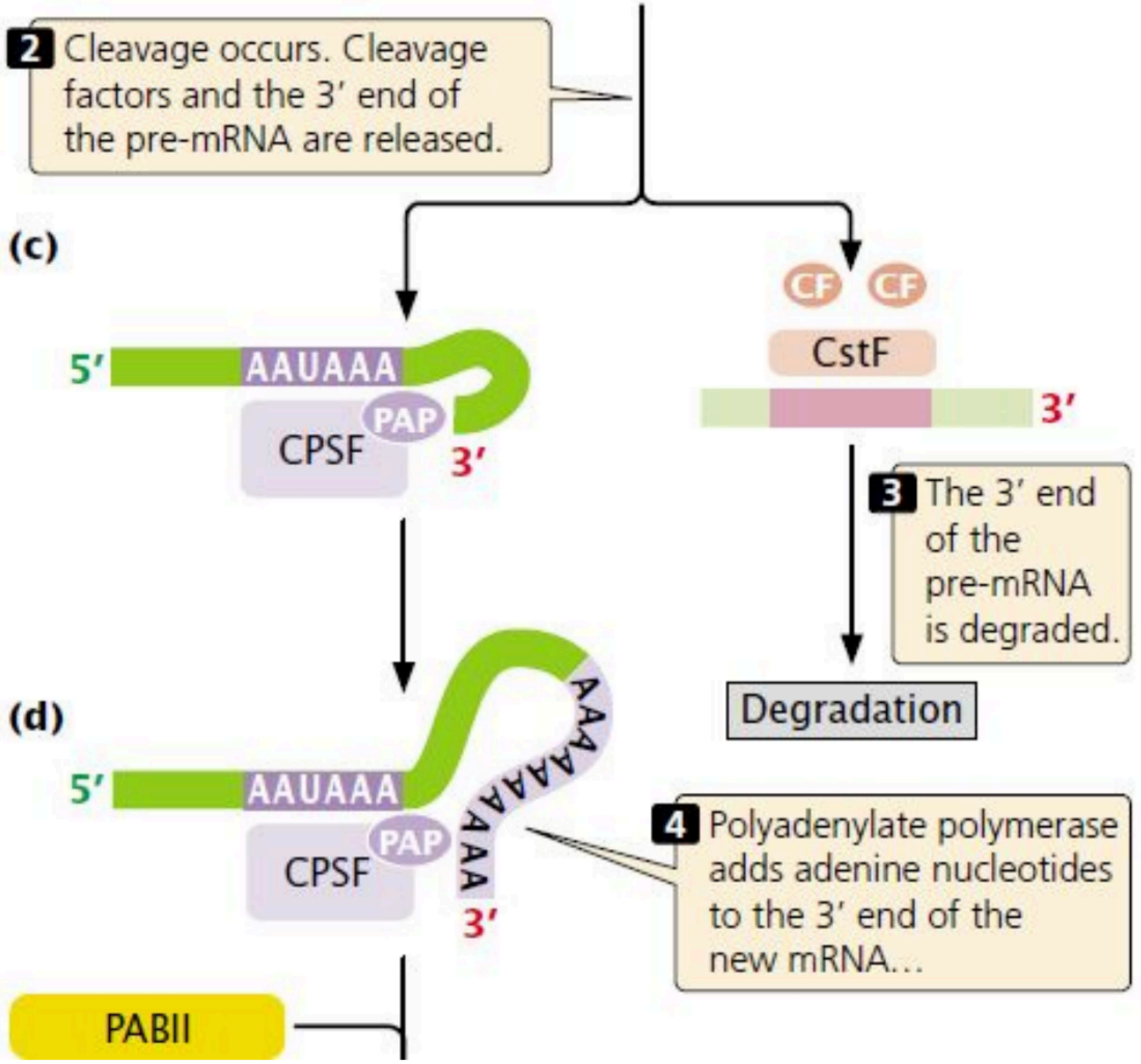
Poly (A) tail

mRNA 5′ AAUAAA AAAAAAAAAAAAAAAAAAAA 3′

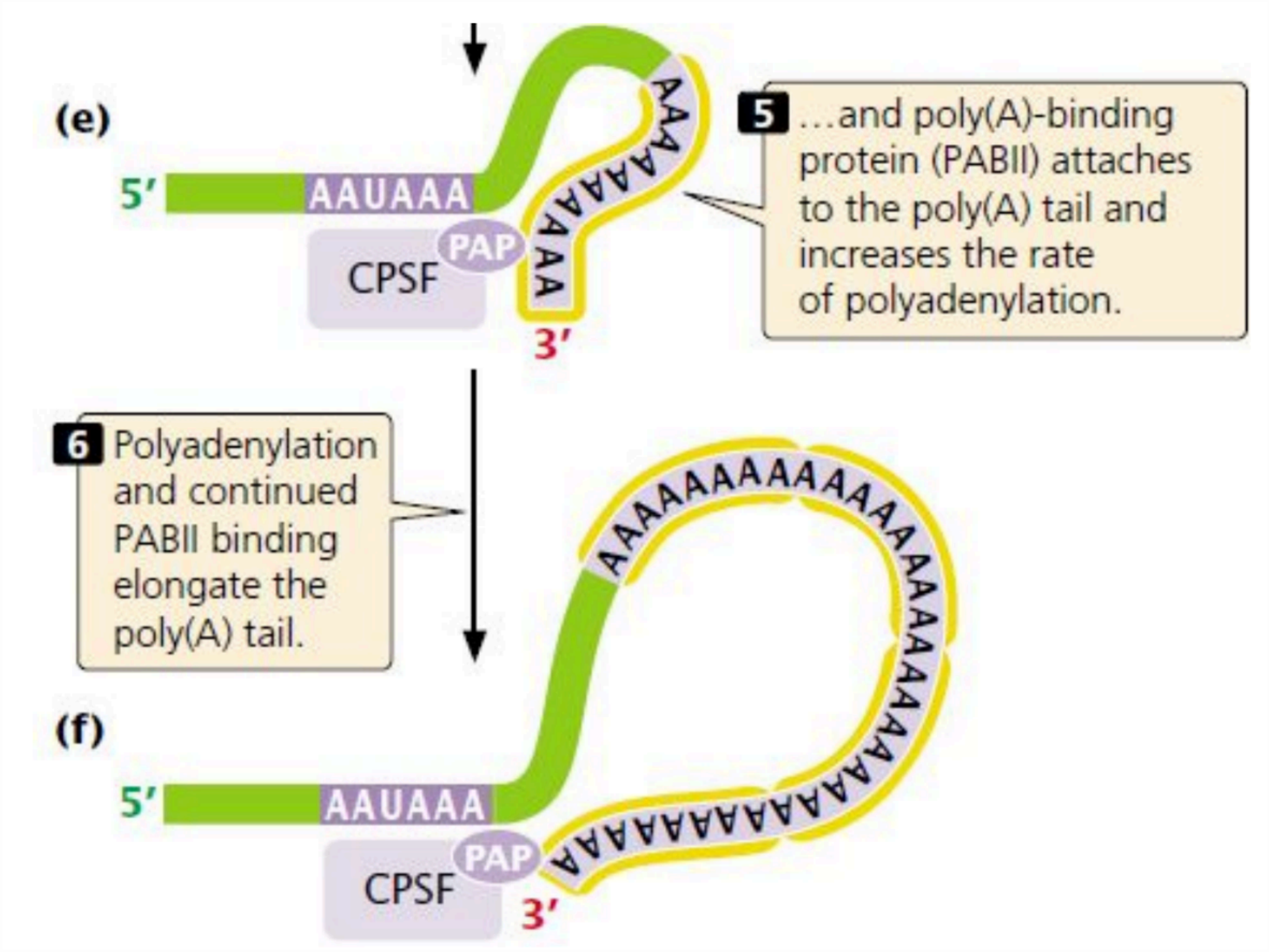
Conclusion: In pre-mRNA processing, a poly(A) tail is added through cleavage and polyadenylation.

- The cleavage site is determined by the consensus sequence located from 11 to 30 upstream and a sequence rich in Us (or Gs and Us) downstream
- Processing of the 3' end of pre-mRNA requires sequences both upstream and downstream of the cleavage site




2 Cleavage occurs. Cleavage factors and the 3′ end of the pre-mRNA are released.
(c)
5′
AAUAAA
PAP
CPSF
3′
CF
CF
CstF
3′
3 The 3′ end of the pre-mRNA is degraded.
Degradation
(d)
5′
AAUAAA
PAP
CPSF
AAAAAAAAA
3′
4 Polyadenylate polymerase adds adenine nucleotides to the 3′ end of the new mRNA…
PABII

The stability of the polyA tail depends on the protein that attach to it



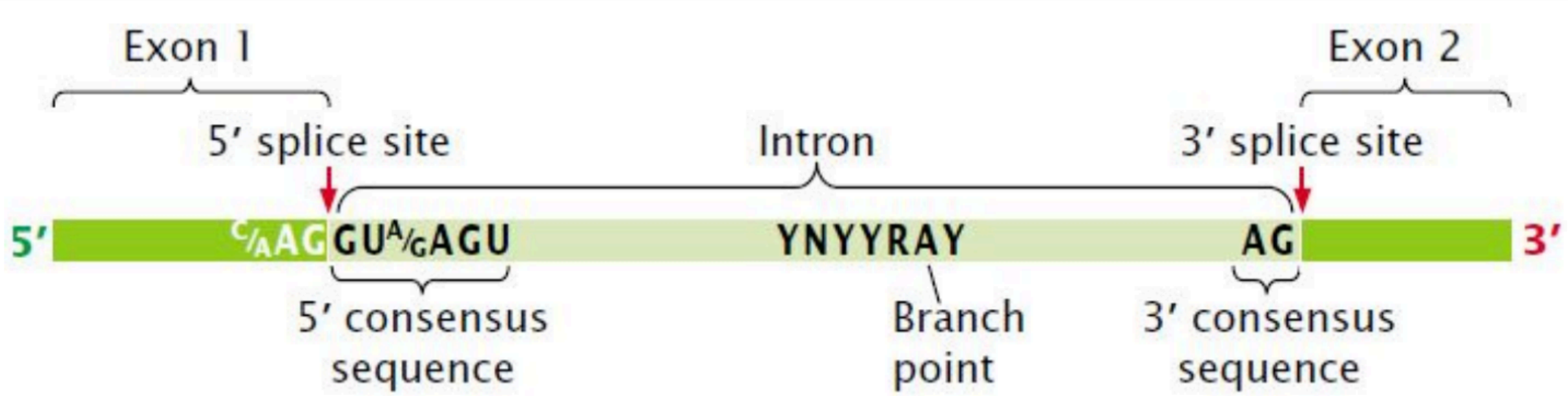
RNA Splicing

Occur in nucleus before RNA is transferred to cytoplasm

Remove introns

Splicing requires 3 sequence in the intron

- 5' splice site: mostly contains GU sequence
- 3' splice site: normally contains AG
- Branch point: nucleotide A that lies between
- 18-40 nucleotide upstream of 3; splice site



- Splicing takes place within the complex called spliceosome
- Pre-mRNA is spliced in to 2 distinct steps
    - Step 1: pre-mRNA is cut at the 5' splice site to free exon 1 and the 5' end of intron attaches to branch point to form lariat
    - Step 2: splicing at 3' end of the intron and join two exons together covalently
- The splicing is actually series of RNA interactions (mRNA-snRNA, snRNA-snRNA...)

Exercise: Which nucleotide at the branch point attacks the 5′ splice site?

A. G

B. U

C. A

D. C

Exercise: Which nucleotide at the branch point attacks the 5′ splice site?

A. G

B. U

C. A

D. C

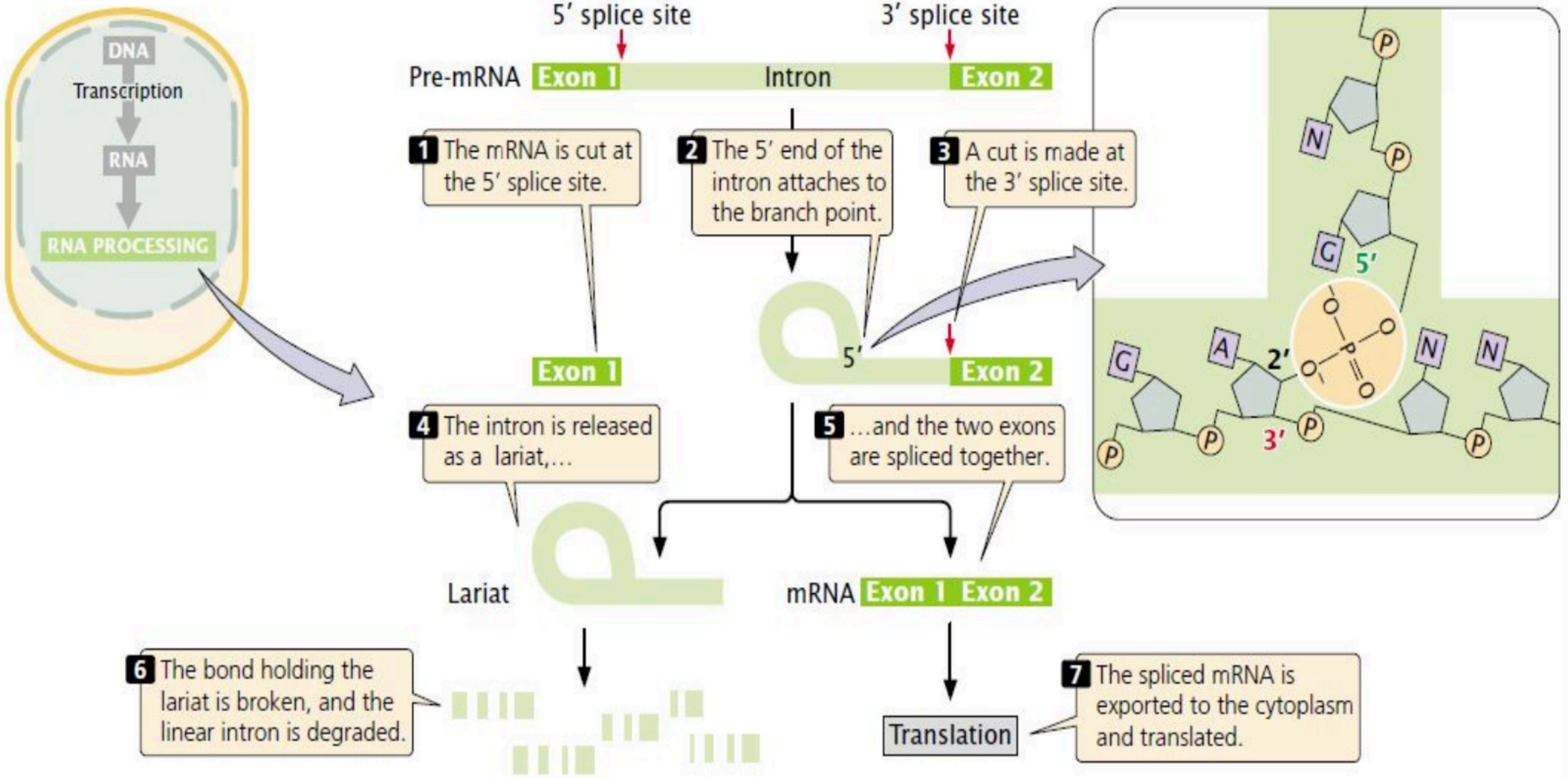
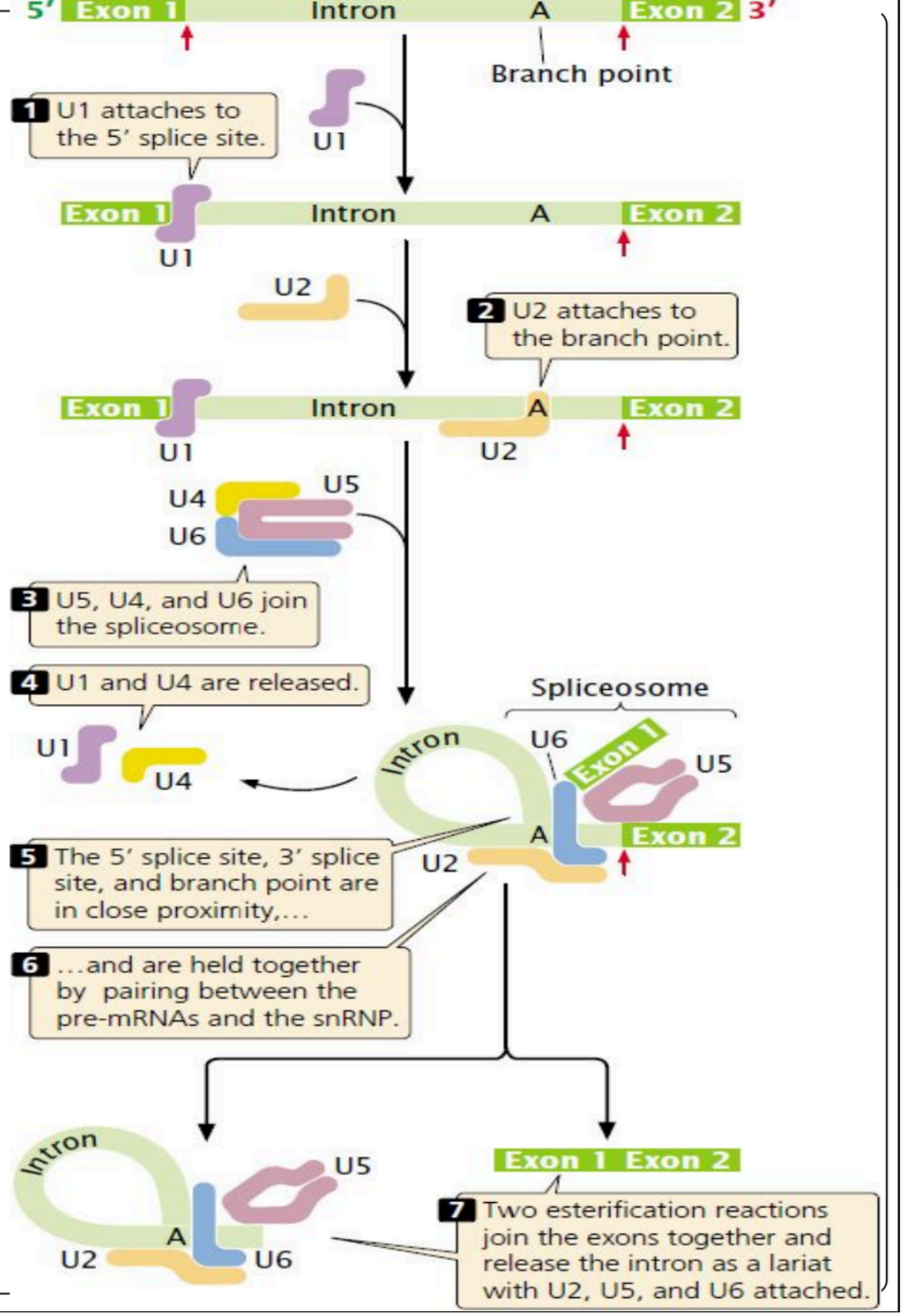

DNA
Transcription
RNA
RNA PROCESSING
5′ splice site
3′ splice site
Pre-mRNA
Exon 1
Intron
Exon 2
1 The mRNA is cut at the 5′ splice site.
2 The 5′ end of the intron attaches to the branch point.
3 A cut is made at the 3′ splice site.
Exon 1
5′
Exon 2
4 The intron is released as a lariat,…
5 …and the two exons are spliced together.
Lariat
mRNA Exon 1 Exon 2
6 The bond holding the lariat is broken, and the linear intron is degraded.
Translation
7 The spliced mRNA is exported to the cytoplasm and translated.
P
N
G
5′
A
2′
3′

Table 14.2 RNA–RNA interactions in pre-mRNA splicing

Interaction	Function
U1 with 5′ splice site	U1 attaches to 5′ end of intron; commits intron to splicing; no direct role in splicing
U2 with branch point	Positions 5′ end of intron near branch point for lariat formation
U2 with U6	Holds 5′ end of intron near branch point
U6 with 5′ splice site	Positions 5′ end of intron near branch point
U5 with 3′ end of first exon	Anchors first exon to spliceosome subsequent to cleavage; juxtaposes two ends of exon for splicing
U5 with 3′ end of one exon and 5′ end of the other	Juxtaposes two ends of exon for splicing
U4 with U6	Delivers U6 to intron; no direct role in splicing



https://www.youtube.com/watch?v=cU60bryCRHM

Self splicing intron

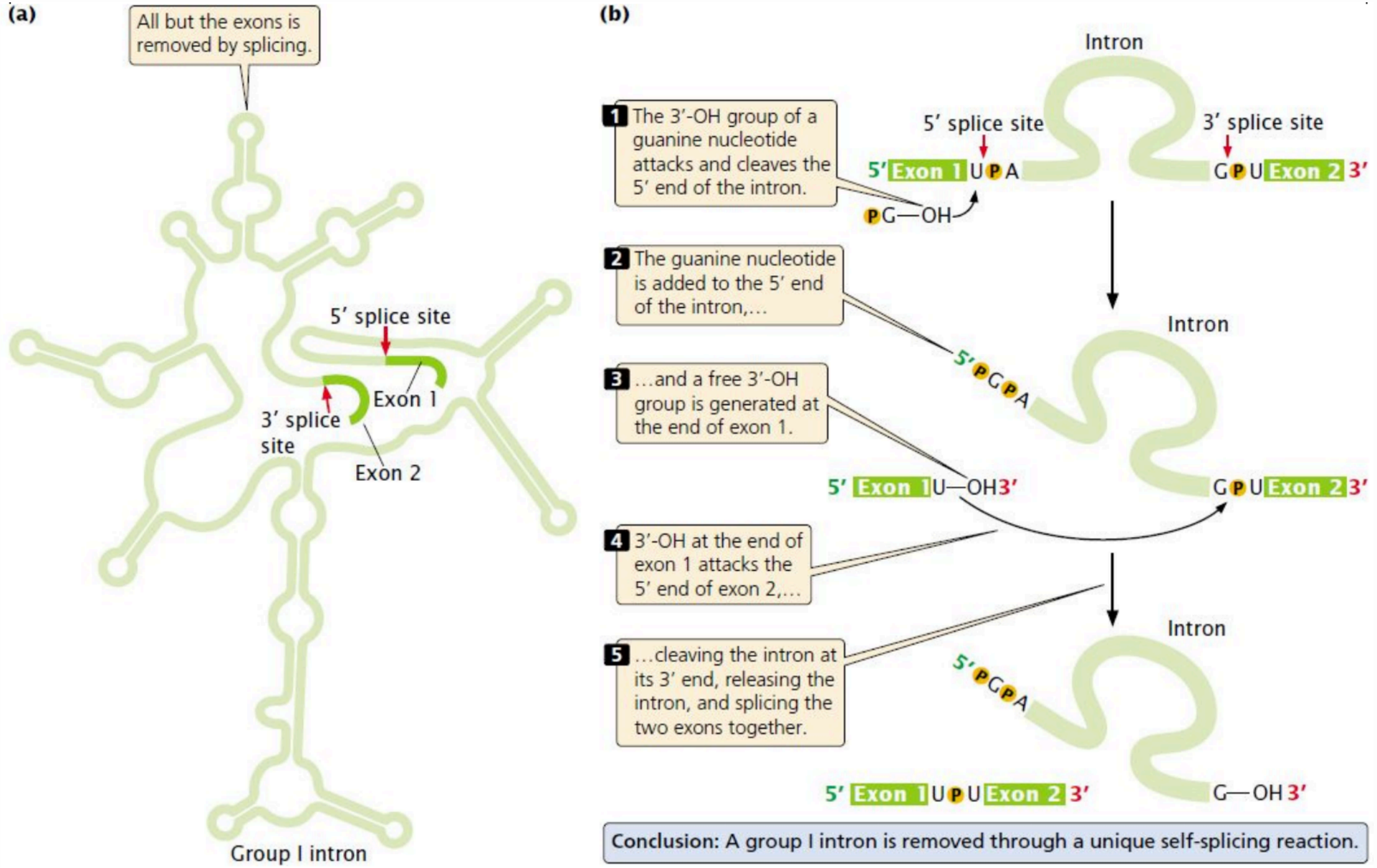
The self splicing introns are able to splice themselves from RNA molecules

Fall into 2 major classes

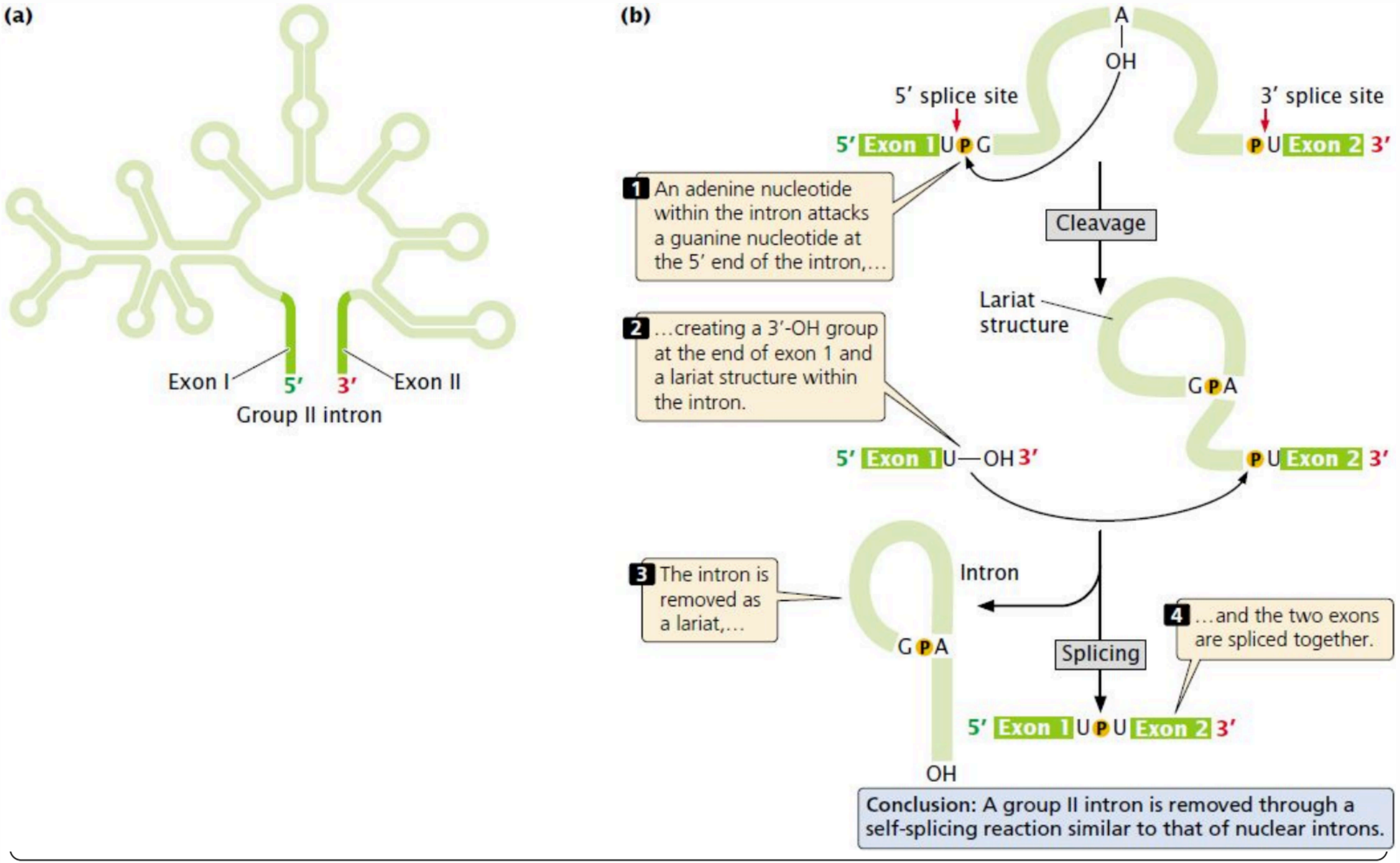
- Group 1: found in variety of genes, fold into common 2nd structure with 9 loop stems
- Group 2: present in some mitochondrion genes

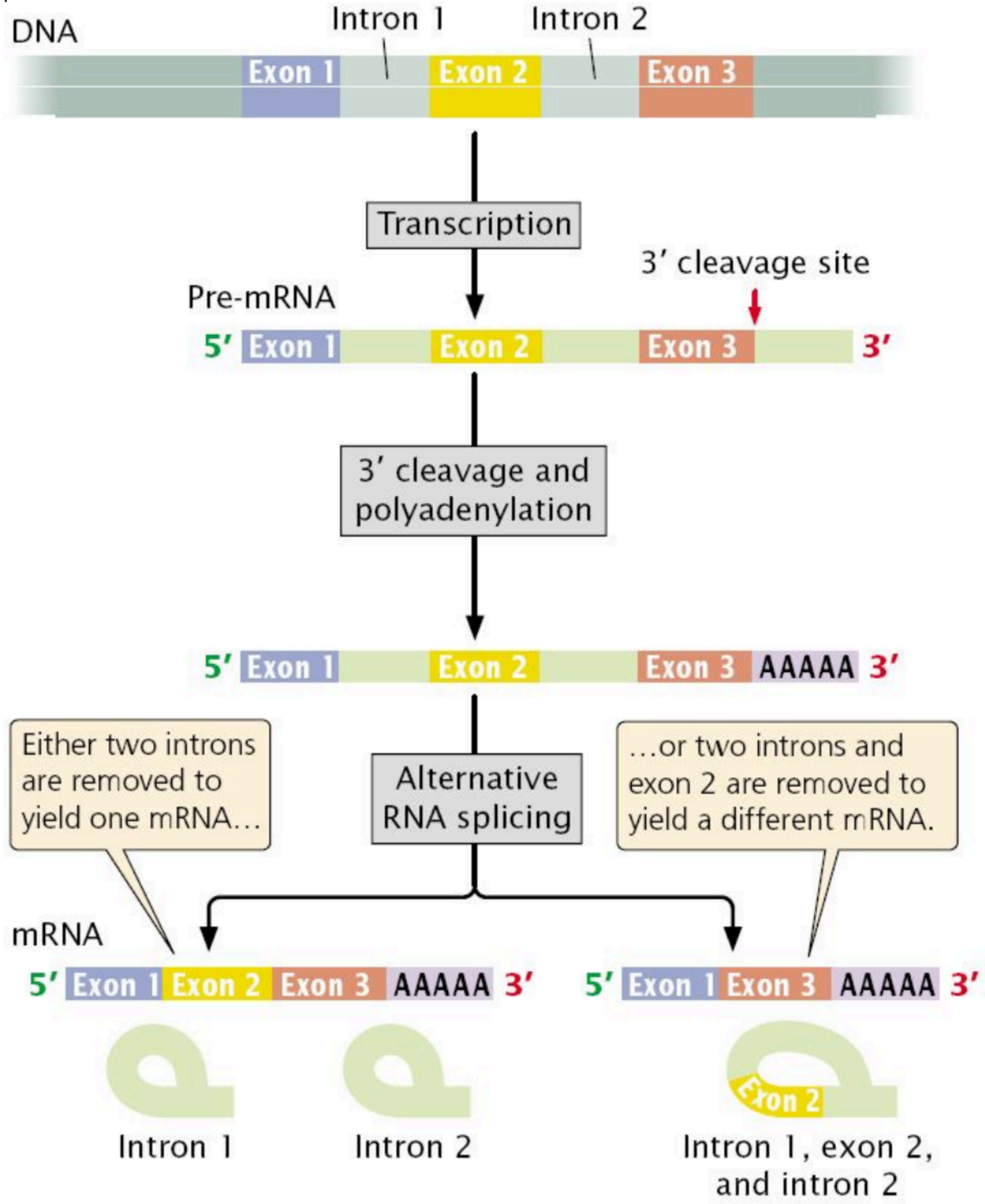
https://www.youtube.com/watch?v=0BOWnqRVLHU&t=167s

Self splicing group I



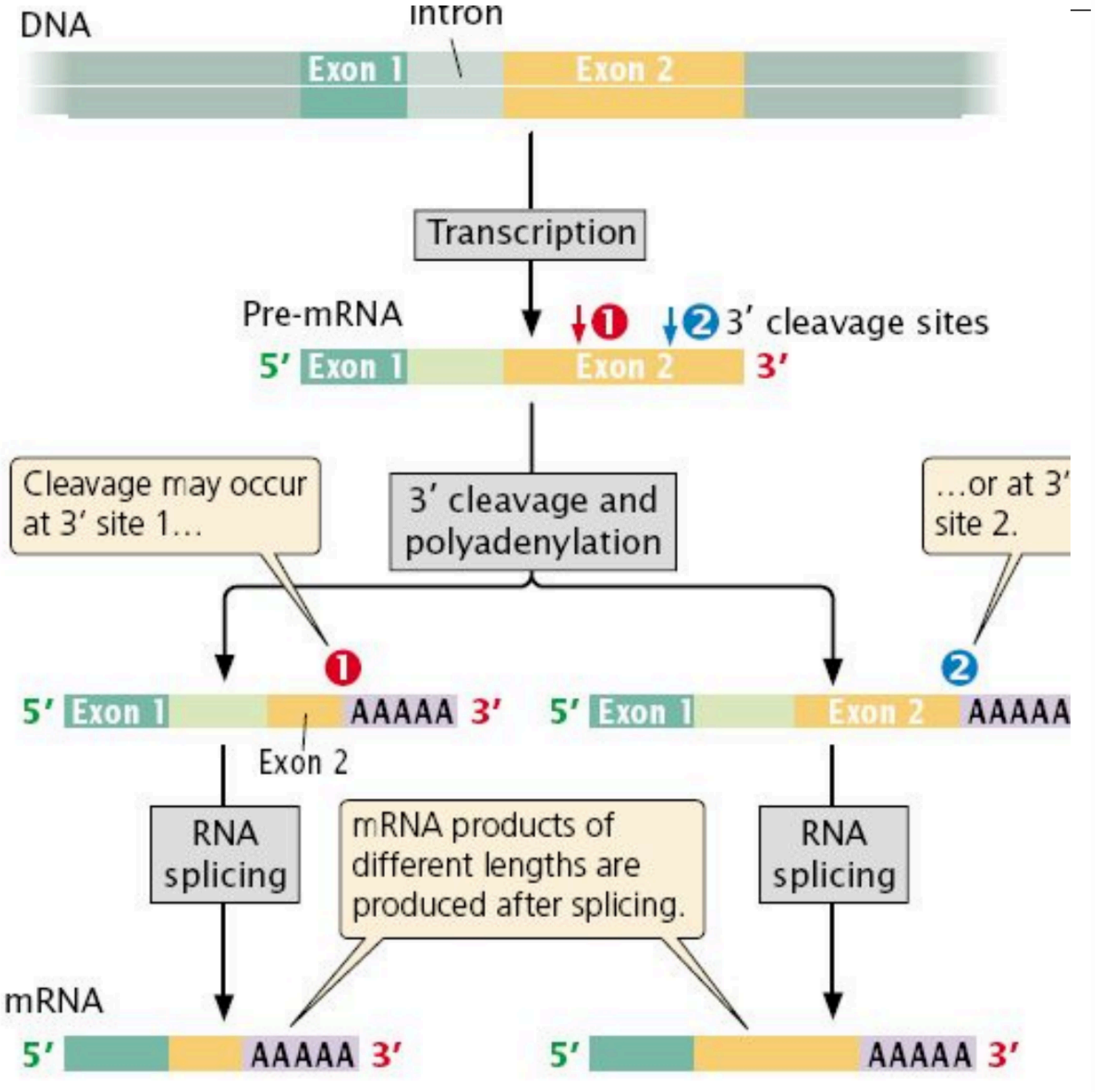
Self splicing group II

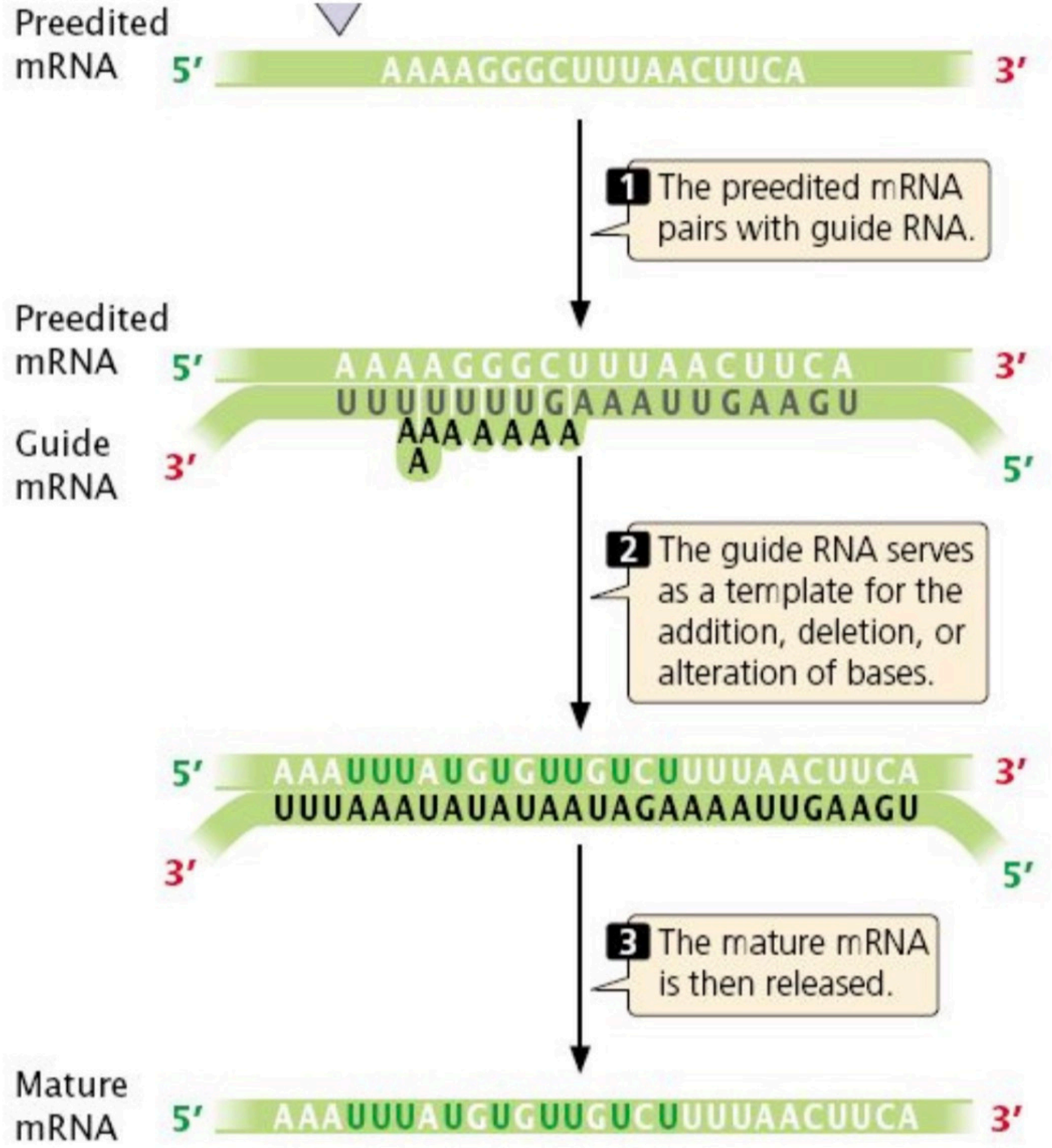




Alternative splicing

same pre-mRNA can be spliced in more than one way to yield multiple mRNAs that are translated into proteins with different amino acid sequences


DNA
Intron
Exon 1
Exon 2
Transcription
Pre-mRNA
1
2
3′ cleavage sites
5′
Exon 1
Exon 2
3′
Cleavage may occur at 3′ site 1…
3′ cleavage and polyadenylation
…or at 3′ site 2.
1
2
5′ Exon 1 AAAAA 3′
Exon 2
5′ Exon 1 Exon 2 AAAAA
RNA splicing
mRNA products of different lengths are produced after splicing.
RNA splicing
mRNA
5′ AAAAA 3′
5′ AAAAA 3′



RNA Editing

the coding sequence of an mRNA molecule is altered after transcription, and so the protein has an amino acid sequence that differs from that encoded by the gene.

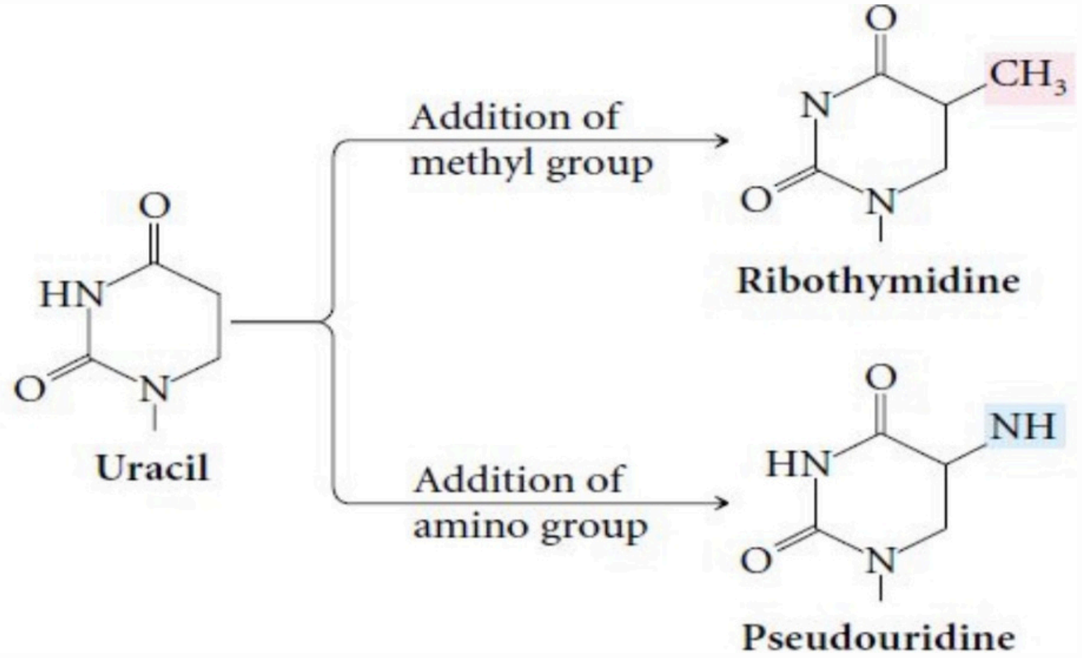
2. Changes in Transfer RNA

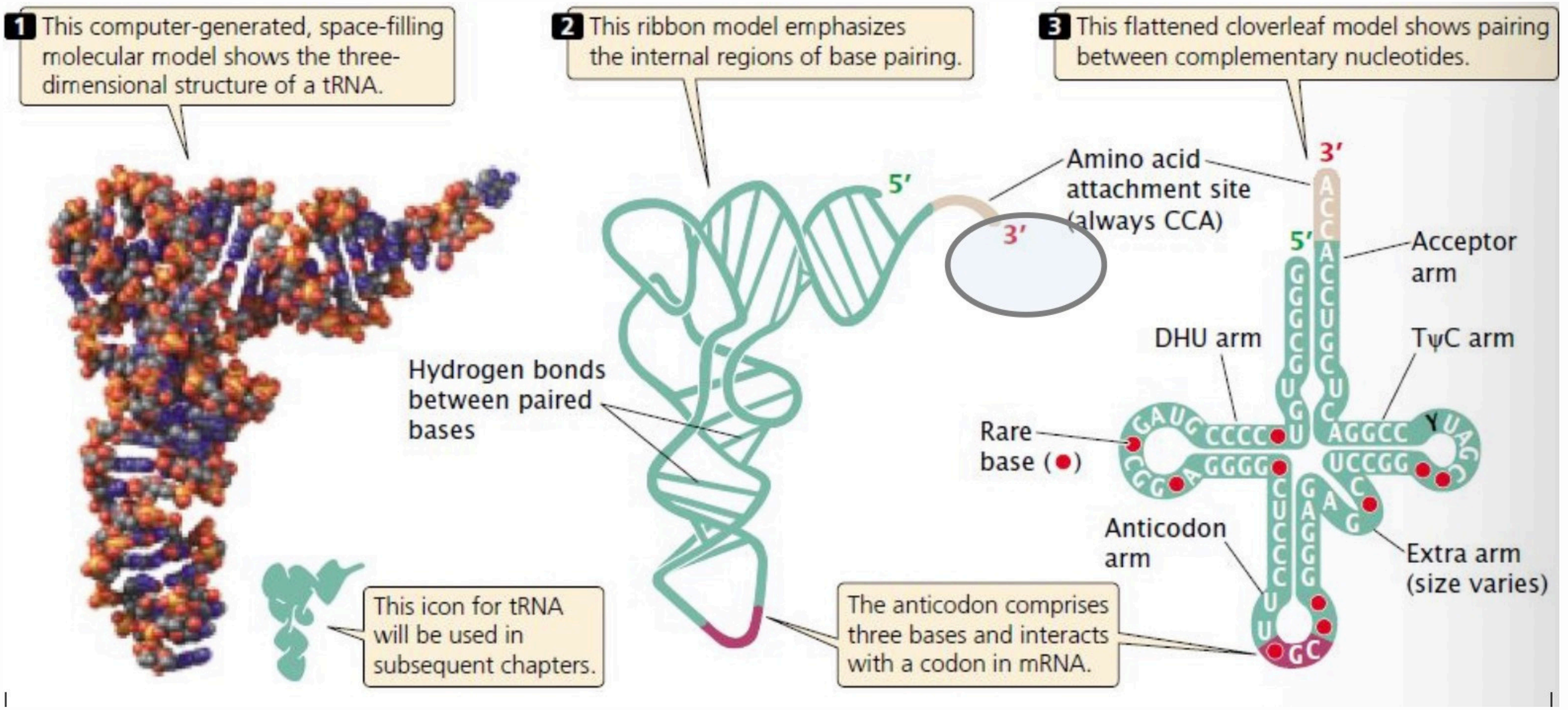
Transfer RNA

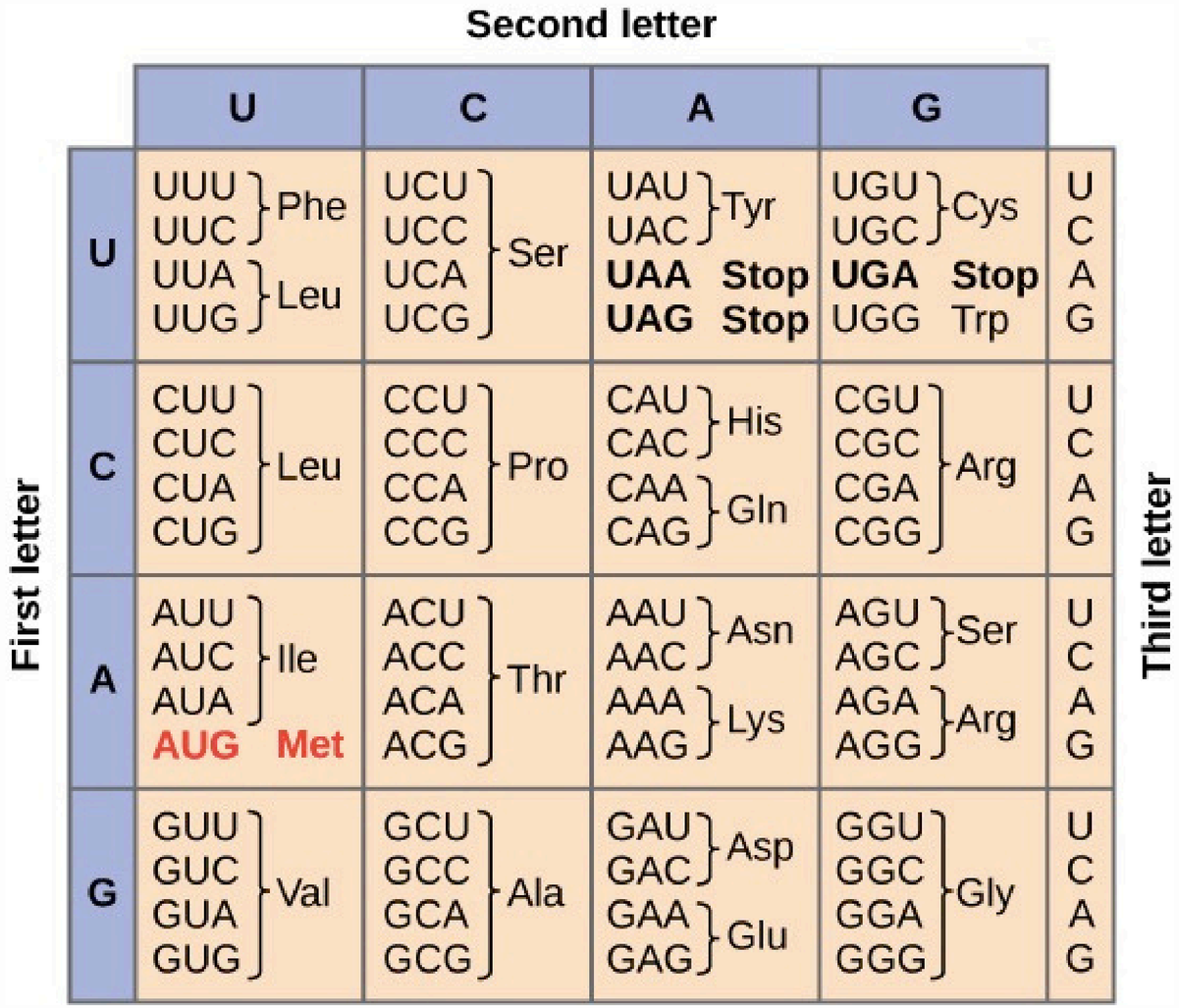
- Serves as link between genetic codes in RNA and amino acids that make up protein
- Each tRNA is capable of attaching to only one type of amino acid
- Most organisms possess from 30 – 40 different types tRNAs
- Each tRNA is encoded by a particular gene, or in some cases, tRNAs are encoded by multiple copies of a gene

Structure of tRNA

- Contains the strange bases which modified from the existed bases after transcription by modifying enzymes
- Similar in 2nd structure
- Contain between 74-95 nucleotides, some of which are complementary to each other to form intramolecular hydrogen bond




1 This computer-generated, space-filling molecular model shows the three-dimensional structure of a tRNA.
2 This ribbon model emphasizes the internal regions of base pairing.
3 This flattened cloverleaf model shows pairing between complementary nucleotides.
5′
3′
Amino acid attachment site (always CCA)
3′
5′
Acceptor arm
DHU arm
TψC arm
Hydrogen bonds between paired bases
Rare base (●)
Anticodon arm
Extra arm (size varies)
This icon for tRNA will be used in subsequent chapters.
The anticodon comprises three bases and interacts with a codon in mRNA.



Codon:

- in mRNA
- total: 64 codon (61 code for amino acid and 3 stopcodon) - 5' → 3' direction

Anticodon:

- in tRNA
- total: ~ 30-40 in most organism

DNA Coding strand: 5' - ATG - 3'

DNA Template: 3' - TAC - 5'

Codon in mRNA:

Anticodon in tRNA:

DNA Coding strand: 5' - ATG - 3'

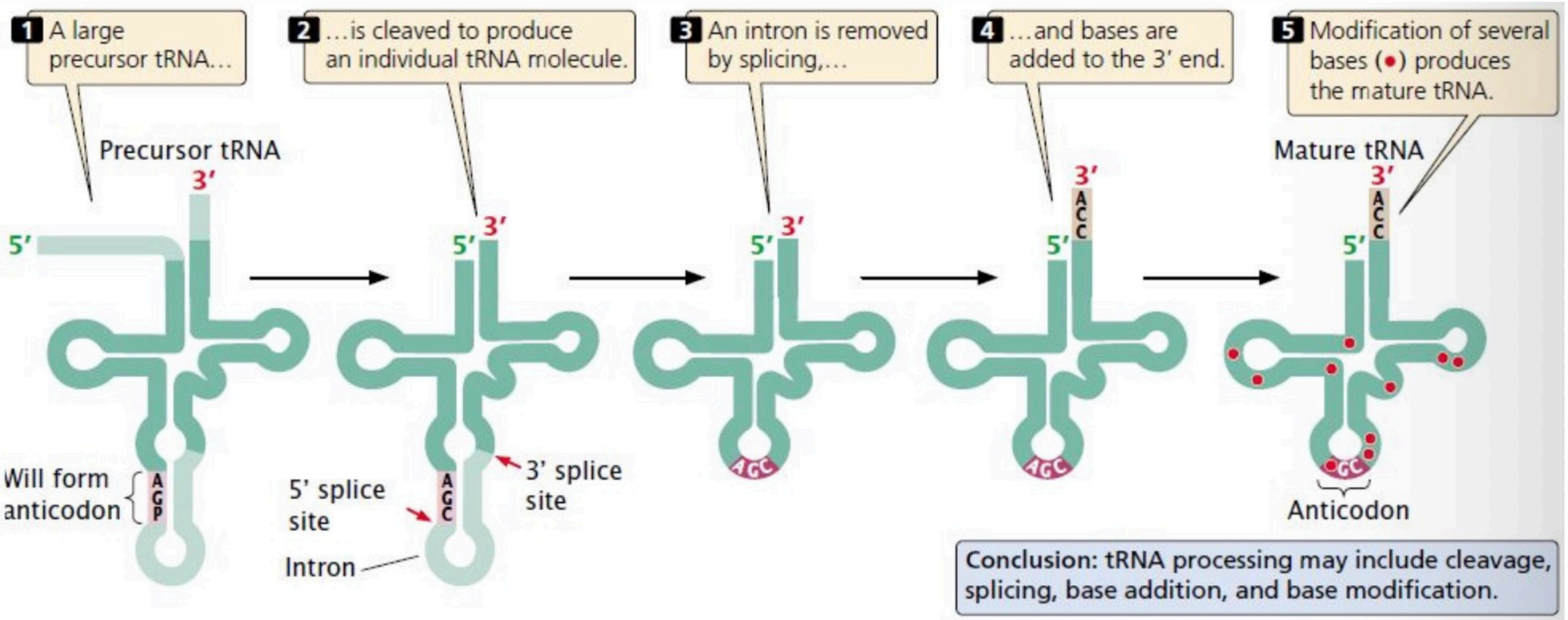
DNA Template: 3' - TAC - 5'

Codon in mRNA: 5' - AUG - 3' (Like Coding strand)

Anticodon in tRNA: 3' - UAC - 5' (Like DNA Template)

tRNA processing

- In E.coli, several tRNAs are transcribed together as large RNA molecule which is then cut into pieces, each contains a single tRNA
- Eukaryotic tRNAs are processed in the same manner by transcription and cutting
- Some eukaryotic tRNAs possess intron that should be spliced


1 A large precursor tRNA…
2 …is cleaved to produce an individual tRNA molecule.
3 An intron is removed by splicing,…
4 …and bases are added to the 3′ end.
5 Modification of several bases (•) produces the mature tRNA.
Precursor tRNA
Mature tRNA
5′
3′
ACC
AGC
Will form anticodon
5′ splice site
3′ splice site
Intron
Anticodon
Conclusion: tRNA processing may include cleavage, splicing, base addition, and base modification.

3. Changes Ribosomal RNAs

Ribosomes are complex organelles, each consists of more than 50 different protein and RNA molecules

Table 14.3 Composition of ribosomes in bacterial and eukaryotic cells

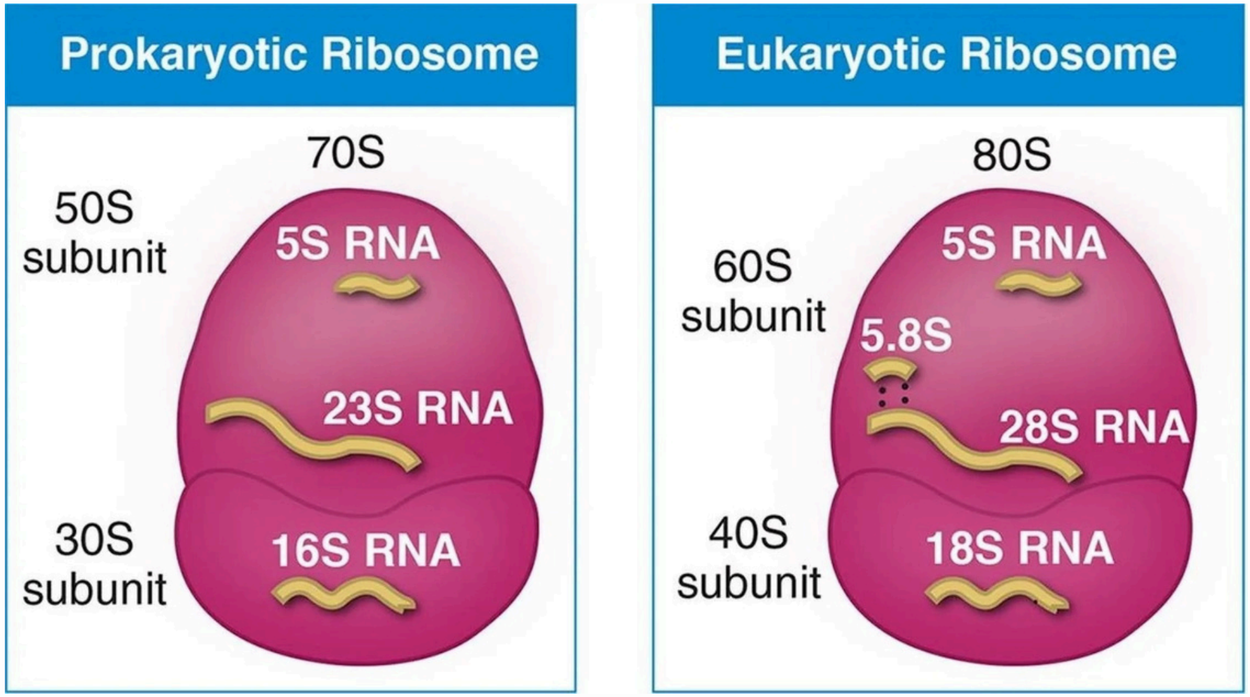
Cell Type	Ribosome Size	Subunit	rRNA Component	Proteins
Bacterial	70S	Large (50S)	23S (2900 nucleotides) 5S (120 nucleotides)	31
		Small (30S)	16S (1500 nucleotides)	21
Eukaryotic	80S	Large (60S)	28S (4700 nucleotides) 5.8S (160 nucleotides) 5S (120 nucleotides)	49
		Small (40S)	18S (1900 nucleotides)	33

Genes for rRNA can be present in multiple copies and number varies among species

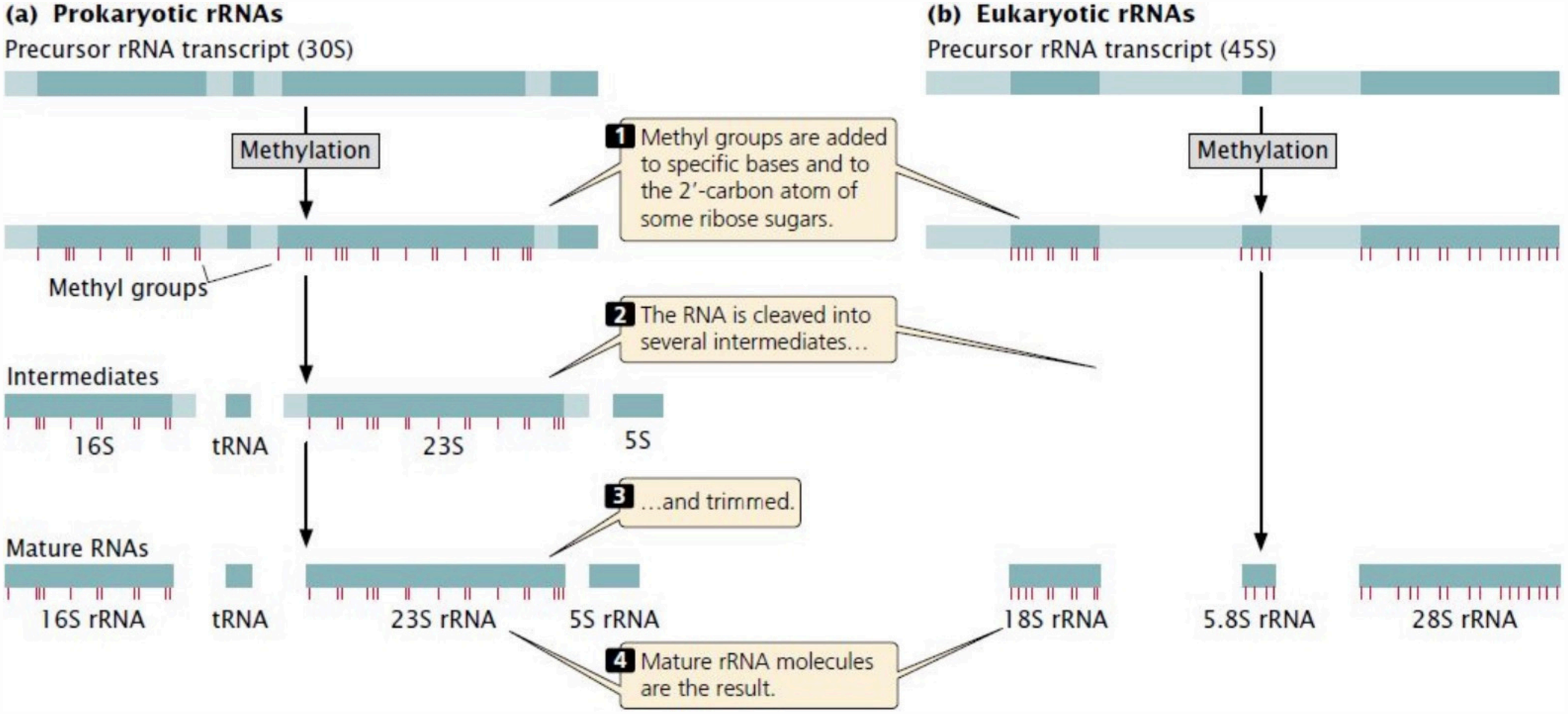
All copies of rRNA genes are identical or nearly identical

In bacteria, genes are dispersed but in eukaryote, genes are clustered

- Eukaryotic cells possess 2 types of rRNA gene: large one encodes 18S, 28S and 5.8S rRNA, small one encodes 5S rRNA
- Bacterial cells contain single gene that codes for all bacterial rRNA: 23S, 16S and 5S rRNA



rRNAs are processed in both bacterial and eukaryotic cells by chemical modification and cleavage



Practice 01:

Codon (mRNA)	Amino acid	Anticodon (tRNA)
5′-GCU-3′	Ala	3′-CGA-5′
5′-UUC-3′	Phe	3′-AAG-5′
5′-CAG-3′	Glu	3′-GUC-5′
5′-UGG-3′	Tryp	3′-ACC-5′

Practice 02:

No.	Term	Letter
1	Exon	G
2	Intron	B
3	5' cap	E
4	Poly(A) tail	C
5	Spliceosome	A
6	Alternative splicing	F
7	RNA editing	D

Definitions

- A. A large complex of snRNAs and proteins that removes introns from pre-mRNA
- B. A sequence in the pre-mRNA that is cut out before the mRNA leaves the nucleus
- C. A chain of about 200–250 adenines added to the 3′ end of mRNA
- D. A change in the sequence of an mRNA after transcription, for example C → U
- E. A 7-methylguanosine joined to the 5′ end of mRNA by a 5′–5′ linkage
- F. Joining exons in different combinations to make several mRNAs from one gene
- G. A sequence that remains in the mature mRNA after splicing

THANK YOU FOR YOUR ATTENTION !